

Original Relation Bulk Control

Exterior density, full Gram estimates and contrast transport

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The complete new proofs retain the original quartet-sum polynomial, all primary multiplicities, both high relation degrees, the original Gamma mass, and the exact coefficient maps. The calculation controls the integration regions omitted from the retained segment and carries the resulting estimates through the full original relation Grams.

The accepted joint-Schur continuation, including its complete typed source and provenance, is supplied as a preserved dependency. This separate reader prints the new analytic calculations. The source acceptance and display-change records identify every file in this edition.

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Complete control of both original high relation degrees

The calculation below retains every root and multiplicity of the original sum packet, its original coefficient, the Gamma source mass, and both high relation degrees. It first proves the density expansion, then transports it through the actual polynomial multiplication map. Finally it bounds both omitted integration regions relative to the retained segment, so that the complete relation determinants have a proved error $o(q)$ in the stated growing family. The resulting explicit integral determinants remain to be evaluated; no value is assigned to them by this comparison.

1 Every original root and its convergent exterior expansion

Fix $0 < \delta < 1/2$, $\gamma > 2$, integers $m, l \geq 1$, and set

$$\begin{aligned} d &= \delta^2, \quad g = \gamma^2, \quad k = 4l + 1, \quad c = k/2, \quad e = 1 + k(m - 1), \quad q = e(k + 1)^2, \\ \omega_{ab} &= (2a - k)\delta + i(2b - k)\gamma, \quad \lambda_{ab} = c + \omega_{ab}, \\ \chi(S) &= \prod_{a,b=0}^k (S - \lambda_{ab})^e, \quad R_* = k\sqrt{d + g}, \quad f(S) = \prod_{j=0}^{l-1} (S - c - t_j), \quad t_j = 2j + 1/2. \end{aligned} \tag{BRD1}$$

Both reflections of each root and its conjugate occur with exactly the same multiplicity. In particular every odd power sum vanishes. Write $P_j = e \sum_{a,b} \omega_{ab}^j$. Conjugation proves P_j is real, and the replacement $(a, b) \mapsto (k - a, k - b)$ proves $P_j = 0$ for odd j . Directly,

$$\begin{aligned} \sum_{a=0}^k (2a - k) &= 0, \quad \sum_{a=0}^k (2a - k)^2 = \frac{(k + 1)k(k + 2)}{3}, \\ P_2 &= \frac{qk(k + 2)(d - g)}{3} =: p_2 < 0. \end{aligned} \tag{BRD2}$$

The second identity follows by expanding $4 \sum a^2 - 4k \sum a + (k + 1)k^2$ with $\sum a = k(k + 1)/2$ and $\sum a^2 = k(k + 1)(2k + 1)/6$. In the double sum for P_2 the mixed term vanishes by the first identity, while the two remaining terms give BRD2 with the complete factor $e(k + 1)^2 = q$.

For real $|y| > R_*$, expand each factor by the absolutely convergent series $\log(1 - z) = -\sum_{n \geq 1} z^n/n$ on $|z| < 1$ and take twice its real part. This defines the real logarithm of the squared modulus irrespective of a phase choice for the product. The odd power sums vanish before any estimate is made. Thus

$$\begin{aligned} \log \frac{|\chi(c + iy)|^2}{|y|^{2q}} &= \sum_{j=1}^{\infty} \frac{(-1)^{j+1} P_{2j}}{j y^{2j}} = \frac{p_2}{y^2} + E_\chi(y), \\ \tau(y) &= R_*^2/y^2 < 1, \quad |E_\chi(y)| \leq q \sum_{j=2}^{\infty} \frac{\tau(y)^j}{j} \leq \frac{q\tau(y)^2}{2(1 - \tau(y))}. \end{aligned} \tag{BRD3}$$

Indeed $|P_{2j}| \leq qR_*^{2j}$. The exact preceding series is also bounded by $q[-\log(1 - \tau) - \tau]$; its geometric bound in BRD3 uses $1/j \leq 1/2$ only for the omitted powers $j \geq 2$. The full polynomial and its first moment coefficient have not been replaced by a freely chosen degree- q polynomial.

At $y = qz$ define

$$a_k = \frac{p_2}{q^2} = \frac{d - g}{3e} \left(1 - \frac{1}{(k + 1)^2} \right), \quad \tau_\epsilon = \frac{k^2(d + g)}{q^2 \epsilon^2}, \quad \Delta_\chi(\epsilon) = \frac{q\tau_\epsilon^2}{2(1 - \tau_\epsilon)}. \tag{BRD4}$$

Whenever $\tau_\epsilon < 1$, BRD3 gives, uniformly on $|z| \geq \epsilon$,

$$|\chi(c + iqz)|^2 = q^{2q} |z|^{2q} \exp\{a_k/z^2 + E_\chi(qz)\}, \quad |E_\chi(qz)| \leq \Delta_\chi(\epsilon). \tag{BRD5}$$

For $m = 1$ the logarithm in BRD5 differs from $(d - g)/(3z^2)$ by at most

$$\frac{g - d}{3(k + 1)^2 \epsilon^2} + \Delta_\chi(\epsilon), \quad (\text{BRD6})$$

which tends to zero as $k \rightarrow \infty$ for each fixed original $d, g, \epsilon > 0$. For fixed $m \geq 2$ its absolute value is at most

$$\frac{g - d}{3\epsilon \epsilon^2} + \Delta_\chi(\epsilon), \quad (\text{BRD7})$$

which also tends to zero. The ratios themselves therefore converge uniformly on $|z| \geq \epsilon$ respectively to $\exp((d - g)/(3z^2))$ and to 1: an absolute logarithmic error b gives a multiplicative interval $[e^{-b}, e^b]$. This retains the actual distinction between the two multiplicity families rather than discarding their factor e .

For the exact Gamma comparison polynomial,

$$\begin{aligned} \frac{|f(c + iy)|^2}{|y|^{2l}} &= \prod_{j=0}^{l-1} (1 + t_j^2/y^2), \quad S_2 := \sum_{j < l} t_j^2 = \frac{l(16l^2 - 12l - 1)}{12}, \\ 0 \leq \log \frac{|f(c + iy)|^2}{|y|^{2l}} &\leq \frac{S_2}{y^2}, \quad -\frac{S_4}{2y^4} \leq \log \frac{|f(c + iy)|^2}{|y|^{2l}} - \frac{S_2}{y^2} \leq 0, \\ S_4 &= \sum_{j < l} t_j^4 = \frac{l(768l^4 - 960l^3 - 160l^2 + 360l + 7)}{240}. \end{aligned} \quad (\text{BRD8})$$

The finite S_2 formula follows by expanding $4j^2 + 2j + 1/4$. For S_4 expand $16j^4 + 16j^3 + 6j^2 + j + 1/16$ and sum. The needed fourth-power sum is $\sum_{j < l} j^4 = l(l - 1)(2l - 1)(3l^2 - 3l - 1)/30$; its value at $l = 0$ is zero and its finite difference at l is l^4 , which proves the formula. The analogous finite differences prove the lower-power sums. Their substitution gives the displayed S_4 without omitting any offset. For $u \geq 0$, integration of $1/(1 + u)$ gives $u - u^2/2 \leq \log(1 + u) \leq u$. Apply these inequalities to each displayed factor to obtain the remainder, with no extra smallness assumption on t_j/y .

2 The original multiplication and coefficient maps

Let $r \geq 1$ and $N = q + r - 1$. Multiplication by the actual monic χ defines

$$T_{\chi,r} : \mathbb{C}[S]_{\leq r-1} \longrightarrow \mathbb{C}[S]_{\leq N}, \quad P \longmapsto \chi P, \quad (T_{\chi,r})_{ij} = \chi_{i-j}. \quad (\text{BRD9})$$

Here χ_h is the coefficient of S^h in the actual polynomial, and is zero outside $0 \leq h \leq q$. Cancellation in $\mathbb{C}[S]$ proves injectivity. Its image is precisely the original relation space $\ker J_N = \chi \mathbb{C}[S]_{\leq r-1}$, by polynomial division. The original relation forms are

$$H_{0,r} = T_{\chi,r}^* H_N^\sigma T_{\chi,r}, \quad H_{f,r} = T_{f,N}^* H_{N+l}^\sigma T_{f,N} \text{ pulled back by } T_{\chi,r}, \quad (\text{BRD10})$$

or equivalently the Grams of χS^j and $f \chi S^j$, $0 \leq j < r$. The notation in the second expression means $H_{f,r} = T_{\chi,r}^* T_{f,N}^* H_{N+l}^\sigma T_{f,N} T_{\chi,r}$. Its measure is the original

$$d\sigma(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi} dy. \quad (\text{BRD11})$$

The coordinate map $\mathcal{T}_r : P(S) \mapsto P(c + iqz)$ has matrix and inverse

$$(\mathcal{T}_r)_{ij} = \binom{j}{i} c^{j-i} (iq)^i \quad (i \leq j), \quad Q(z) \longmapsto Q((S - c)/(iq)), \quad \det \mathcal{T}_r = (iq)^{r(r-1)/2}. \quad (\text{BRD12})$$

The exact intertwining square is $\mathcal{T}_{N+1}T_{\chi,r} = T_{\chi(c+iqz),r}\mathcal{T}_r$. Both sides send P to the very same polynomial $\chi(c+iqz)P(c+iqz)$. This proves the equality of source maps before any integration or determinant comparison.

Fix $0 < \epsilon < R < \infty$ with $q\epsilon > R_*$ and partition the original real line into

$$I = \{|y| < q\epsilon\}, \quad B = \{q\epsilon \leq |y| \leq qR\}, \quad F = \{|y| > qR\}. \quad (\text{BRD13})$$

Let $H_{a,r}^I, H_{a,r}^B, H_{a,r}^F$, for $a = 0, f$, denote the exact restrictions of BRD10 to these regions. Then

$$H_{a,r} = H_{a,r}^I + H_{a,r}^B + H_{a,r}^F. \quad (\text{BRD14})$$

All three forms are positive definite: their scalar weight is positive almost everywhere on a nonempty interval, and a nonzero polynomial cannot vanish on that interval. This also proves that every inverse and determinant below exists.

3 The two full-degree bulk determinant comparisons

For $s = 0, l$ let $K_{s,r}$ be the moment Gram on the original segment in the real z coefficient frame:

$$(K_{s,r})_{ij} = \int_{\epsilon \leq |z| \leq R} z^{i+j} |z|^{2q+2s} \exp(a_k/z^2) \sigma(qz) dz, \quad 0 \leq i, j < r. \quad (\text{BRD15})$$

The corresponding forms in the original S frame are

$$B_{s,r} = q^{2q+2s+1} \mathcal{T}_r^* K_{s,r} \mathcal{T}_r. \quad (\text{BRD16})$$

This retains the Jacobian $dy = q dz$, the density factor q^{2q+2s} , the exact coefficient map, and the original Gamma density $\sigma(qz)$. In particular the source has not been replaced by a probability measure.

Put $\Delta_f(\epsilon) = S_2/(q^2\epsilon^2)$ and $\Delta_\chi = \Delta_\chi(\epsilon)$. The pointwise inequalities BRD5 and BRD8 prove, by integrating each actual squared polynomial, the Loewner inequalities

$$\begin{aligned} e^{-\Delta_\chi} B_{0,r} &\preceq H_{0,r}^B \preceq e^{\Delta_\chi} B_{0,r}, \\ e^{-\Delta_\chi} B_{l,r} &\preceq H_{f,r}^B \preceq e^{\Delta_\chi + \Delta_f(\epsilon)} B_{l,r}. \end{aligned} \quad (\text{BRD17})$$

They have been proved on exactly the stipulated segment B . If $aA \preceq H \preceq bA$ with $A, H > 0$, congruence by $A^{-1/2}$ puts every eigenvalue in $[a, b]$; their product proves $r \log a \leq \log \det H - \log \det A \leq r \log b$. Consequently

$$\begin{aligned} \log \frac{\det H_{f,r}^B}{\det H_{0,r}^B} &= 2lr \log q + \log \frac{\det K_{l,r}}{\det K_{0,r}} + b_r, \\ -2r\Delta_\chi &\leq b_r \leq 2r\Delta_\chi + r\Delta_f(\epsilon). \end{aligned} \quad (\text{BRD18})$$

The determinant of $\mathcal{T}_r$ cancels only in this ratio: each individual determinant in BRD16 has factor $q^{r(2q+2s+1)+r(r-1)}$.

For fixed d, g, m, ϵ and $r = q$ or $q + 1$, BRD4 gives the explicit finite error

$$2r\Delta_\chi + r\Delta_f(\epsilon) = \frac{rk^4(d+g)^2}{q^3\epsilon^4(1-\tau_\epsilon)} + \frac{rS_2}{q^2\epsilon^2}. \quad (\text{BRD19})$$

Its first term is $O(1)$ for $m = 1$ and $O(k^{-2})$ for fixed $m \geq 2$. Its second term is $O(k)$ and $O(1)$, respectively. Indeed $r/q \leq 1 + 1/q$, $q = e(k+1)^2$, $S_2 \leq 4l^3$, and $e = 1$ or $e = 1 + k(m-1)$; substituting these actual degrees gives each stated bound. In both cases BRD19 is $o(q)$. A bound on a density alone would not establish this: the full relation dimension r is explicitly present in BRD19.

4 Exact tail determinants before any estimate

Define the actual, dimension- r positive matrix

$$D_{a,r} = (H_{a,r}^B)^{-1/2} (H_{a,r}^I + H_{a,r}^F) (H_{a,r}^B)^{-1/2}, \quad t_{a,r} = \log \det(I_r + D_{a,r}), \quad a = 0, f. \quad (\text{BRD20})$$

Factoring BRD14 by the two square roots of its bulk Gram gives the exact identities

$$\begin{aligned} \log \det H_{a,r} &= \log \det H_{a,r}^B + t_{a,r}, \\ \log \Re_{q+r-1} &= 2lr \log q + \log \frac{\det K_{l,r}}{\det K_{0,r}} + b_r + t_{f,r} - t_{0,r}, \quad \Re_N = \det H_{f,r} / \det H_{0,r}. \end{aligned} \quad (\text{BRD21})$$

The same determinant equals $\det[I + (H_{a,r}^B)^{-1} (H_{a,r}^I + H_{a,r}^F)]$ by similarity. No tail is removed by this equality. The following calculation now proves an actual bound for both of these exact corrections.

5 An explicit common source estimate for the inner and far regions

Set once and for all $\epsilon = 2^{-10}$ and $R = 64$. For the uniform statements in this section impose the explicit finite thresholds

$$\frac{k\sqrt{d+g}}{q} \leq \epsilon/2, \quad \frac{2l-3/2}{q} \leq \epsilon. \quad (\text{BRD22})$$

Both hold eventually for every fixed original d, g, m . For example $k \geq 2048\sqrt{d+g}$ suffices: $q \geq k^2$ proves the first inequality, and $t_{l-1} < k/2$ proves the second because $\sqrt{d+g} > 2$. In particular $\tau_\epsilon \leq 1/4$, so the bulk expansion has a strict convergence margin even at the displayed threshold. These inequalities specify which already defined finite sources satisfy the uniform bounds; they assert no property of a zeta zero. For $r = q$ or $q+1$, write $Q(z) = P(c+iqz)$, so $\deg Q \leq r-1 \leq q$. The positive subsegment $1 \leq z \leq 2$ lies inside B .

Here are the needed Gamma bounds with their original constants. Rotate the Euler integral of $\Gamma(a+iy/2)$ through the angle $\text{sgn}(y)\theta$, where $0 < \theta < \pi/2$. On the principal branch its value is

$$e^{i \text{sgn}(y)\theta(a+iy/2)} \int_0^\infty t^{a+iy/2-1} e^{-te^{i \text{sgn}(y)\theta}} dt.$$

The circular arcs vanish at infinity by positive cosine and at zero by $a > 0$, so Cauchy's theorem proves the contour identity. Taking absolute values proves $|\Gamma(a+iy/2)| \leq e^{-\theta|y|/2} \Gamma(a)(\cos \theta)^{-a}$. At $a = 1/4$, $\theta = 1$, this gives

$$\sigma(y) \leq C_U e^{-|y|}, \quad C_U = \frac{\Gamma(1/4)^2}{2\pi\sqrt{\cos 1}}. \quad (\text{BRD23})$$

Here is the needed reflection identity on its precise domain. For $0 < \text{Re } z < 1$, the product of the two absolutely convergent Euler integrals, with $u = tv$, gives

$$\Gamma(z)\Gamma(1-z) = \int_0^\infty \frac{t^{z-1}}{1+t} dt.$$

Integrate $w^{z-1}/(1+w)$ around a keyhole contour about the positive real axis with argument in $(0, 2\pi)$. The large and small circular contributions are respectively $O(R^{\text{Re } z-1})$ and $O(\rho^{\text{Re } z})$, so both vanish. The two straight segments give $(1 - e^{2\pi iz})$ times the displayed real integral. The only pole is $w = -1$, with residue $e^{i\pi(z-1)}$. The residue theorem therefore evaluates the real integral as $2\pi i e^{i\pi(z-1)}/(1 - e^{2\pi iz}) = \pi/\sin(\pi z)$. This proves the reflection identity used here, including at $z = 1/4 + iy/2$. At that point it gives $|\Gamma(1/4 + iy/2)|^2 |\Gamma(3/4 - iy/2)|^2 = 2\pi^2/\cosh(\pi y)$. The unrotated Euler integral bounds the second Gamma factor by $\Gamma(3/4)^2$, whence

$$\sigma(y) \geq \frac{C_L}{\cosh(\pi y)} \geq C_L e^{-\pi|y|}, \quad C_L = \frac{\pi}{\Gamma(3/4)^2}, \quad \frac{C_U}{C_L} = \frac{1}{\sqrt{\cos 1}}. \quad (\text{BRD24})$$

The last equality uses the reflection identity at $1/4$, $\Gamma(1/4)\Gamma(3/4) = \pi\sqrt{2}$.

For the polynomial bound define the Legendre polynomials by $P_j(x) = (2^j j!)^{-1} (d/dx)^j (x^2 - 1)^j$. Their leading coefficient is $a_j = (2j)!/[2^j (j!)^2]$. Every derivative of $(x^2 - 1)^j$ of order less than j vanishes at both endpoints. Thus j integrations by parts prove orthogonality against lower degrees, and for the norm they give explicitly

$$\int_{-1}^1 P_j(x)^2 dx = \frac{a_j}{2^j} \int_{-1}^1 (1 - x^2)^j dx = \frac{a_j}{2^j} \frac{2^{2j+1} (j!)^2}{(2j+1)!} = \frac{2}{2j+1}.$$

The middle integral follows by $x = 2t - 1$ and the beta integral $\int_0^1 t^j (1-t)^j dt = (j!)^2/(2j+1)!$, which also follows by repeated integration by parts. Consequently $\sqrt{2j+1} P_j(2z-3)$, $0 \leq j < r$, are an orthonormal basis on $[1, 2]$ for the polynomial space in question. For real $|x| \geq 1$, their integral representation

$$P_j(x) = \frac{1}{\pi} \int_0^\pi (x + \sqrt{x^2 - 1} \cos \theta)^j d\theta \quad (\text{BRD25})$$

gives $|P_j(x)| \leq (|x| + \sqrt{x^2 - 1})^j \leq (2|x|)^j$. For negative x use $P_j(-x) = (-1)^j P_j(x)$. The representation follows by expanding the integrand: odd powers of cosine integrate to zero and $\pi^{-1} \int_0^\pi \cos^{2h} \theta d\theta = \binom{2h}{h}/4^h$, obtained by integration by parts. The resulting polynomial is the Rodrigues polynomial, as direct expansion of its derivative verifies. Thus the bound uses an explicit polynomial identity. Expansion of Q in this orthonormal basis and Cauchy–Schwarz therefore proves

$$|Q(z)|^2 \leq \sum_{j=0}^{r-1} (2j+1) |P_j(2z-3)|^2 \int_1^2 |Q(t)|^2 dt. \quad (\text{BRD26})$$

Pair every root ω with $-\omega$, keeping multiplicities. For real y , $|(-iy - \omega)(-iy + \omega)| \leq y^2 + |\omega|^2$, and for $|y| > R_*$ the reverse triangle inequality gives the lower bound $y^2 - R_*^2$. Consequently BRD22 implies on $1 \leq z \leq 2$

$$|\chi(c + iqz)|^2 \geq q^{2q} (3/4)^q, \quad |f(c + iqz)|^2 \geq q^{2l}. \quad (\text{BRD27})$$

In particular, for $a = 0, f$ and $s = 0, l$ respectively, the actual bulk quadratic form is at least

$$q^{2q+2s+1} (3/4)^q C_L e^{-2\pi q} \int_1^2 |Q(z)|^2 dz. \quad (\text{BRD28})$$

On $|z| \leq \epsilon$, the upper paired-root estimate and BRD22 give

$$|\chi(c + iqz)|^2 \leq q^{2q} (2\epsilon^2)^q, \quad |f(c + iqz)|^2 \leq q^{2l} (2\epsilon^2)^l, \quad \sum_{j < r} (2j+1) |P_j(2z-3)|^2 \leq r^2 8^{2q}. \quad (\text{BRD29})$$

For the last bound $|2z-3| \leq 3 + 2\epsilon$, so BRD25 bounds each polynomial by $(6 + 4\epsilon)^j \leq 8^j$, and $\sum_{j < r} (2j+1) = r^2$, $r-1 \leq q$. Integrate over a segment of length 2ϵ , apply BRD23 and BRD26, and divide by BRD28. For both actual forms this proves

$$H_{a,r}^I \preceq \kappa_{I,r} H_{a,r}^B, \quad \kappa_{I,r} = \frac{2\epsilon r^2}{\sqrt{\cos 1}} \left(\frac{4}{3} 2^{-13} e^{2\pi} \right)^q < \frac{2\epsilon r^2}{\sqrt{\cos 1}} e^{-2q}. \quad (\text{BRD30})$$

For the numerator the intermediate bound contains the additional factor $(2\epsilon^2)^l \leq 1$, which was retained in BRD29 before obtaining the common bound. To justify the last strict inequality without a decimal estimate, $\log 2 > 2/3$ follows from the positive series for $2 \operatorname{arctanh}(1/3)$, and $\log(4/3) < 1/3$ follows by integrating $1/t < 1$ on $1 < t < 4/3$. Also

$$\frac{22}{7} - \pi = \int_0^1 \frac{x^4 (1-x)^4}{1+x^2} dx > 0.$$

Polynomial division and $\int_0^1 (1+x^2)^{-1} dx = \pi/4$ prove this identity. Thus the logarithm of the parenthesized base is less than $-26/3 + 44/7 + 1/3 = -43/21 < -2$.

For the far region a sharper use of BRD5 avoids a large coefficient bound. Define $\Delta_\chi(B)$ by BRD4 with ϵ replaced by B , and let

$$\eta = -a_k + \Delta_\chi(1), \quad D_s = 2q + 2s + 2r - 2, \quad \Delta_{f,0}(B) = 0, \quad \Delta_{f,l}(B) = S_2/(q^2 B^2). \quad (\text{BRD31})$$

On $[1, 2]$ one may replace $(3/4)^q$ in BRD28 by $e^{-\eta}$, by BRD5 and $a_k < 0$. On $|z| \geq B \geq 2$, BRD25–26 give the polynomial bound $r^2((4 + 6/B)|z|)^{2r-2} \int_1^2 |Q|^2$. The exact upper weights are at most $q^{2q+2s}|z|^{2q+2s} \exp(\Delta_\chi(B) + \Delta_{f,s}(B))$, because the negative term a_k/z^2 only decreases the bound. For this parameterized estimate define the exact moving far region $F_B = \{|y| > qB\}$ and let $H_{a,r}^{F_B}$ be the original form integrated over that region. The bulk denominator continues to be the fixed region $\epsilon \leq |z| \leq 64$, containing $[1, 2]$. After BRD23 and $dy = q dz$, division by the improved BRD28 therefore gives

$$H_{a,r}^{F_B} \preceq \kappa_{F,s,r} H_{a,r}^B, \quad (\text{BRD32})$$

where at any $B > D_s/q$, $B \geq 2$,

$$\kappa_{F,s,r} = \frac{2r^2}{\sqrt{\cos 1}} \frac{\exp(2\pi q + \eta + \Delta_\chi(B) + \Delta_{f,s}(B))(4 + 6/B)^{2r-2} B^{D_s} e^{-qB}}{q - D_s/B}. \quad (\text{BRD33})$$

The last integral bound used here is fully explicit: $\log(t/B) \leq (t - B)/B$ for $t \geq B$ implies

$$\int_B^\infty t^{D_s} e^{-qt} dt \leq \frac{B^{D_s} e^{-qB}}{q - D_s/B}. \quad (\text{BRD34})$$

It is applied to both tails, which accounts for the factor two in BRD33.

Specialize now to $B = 64$, so $F_B = F$ is exactly the far region of the original partition BRD13. We have $D_s \leq 5q$, since $r - 1 \leq q$ and $2l \leq q$. Define

$$c_F = 64 - 2\pi - 5 \log 64 - 2 \log(131/32) > 20. \quad (\text{BRD35})$$

Indeed $\pi < 4$, $\log 2 < 1$, and $131/32 < 8$ give respectively $2\pi < 8$, $5 \log 64 < 30$, and $2 \log(131/32) < 6$. The weaker threshold $q \geq 2R_*$, already implied by BRD22, gives $\tau_1 \leq 1/4$. Since $k \geq 5$,

$$-a_k \leq 7q/60, \quad \Delta_\chi(1) \leq q/24, \quad \Delta_\chi(64) \leq q/24, \quad \Delta_{f,l}(64) \leq q/8192. \quad (\text{BRD36})$$

For the first inequality use $-a_k \leq q\tau_1(k+2)/(3k)$ and $(k+2)/k \leq 7/5$. For the last, $S_2 \leq 4l^3 \leq q^3/2$ follows from $2l \leq q$. Their sum is less than $q/4$. Hence BRD33 gives the uniform bound, for both $s = 0, l$,

$$\kappa_{F,s,r} \leq \frac{128r^2}{59q\sqrt{\cos 1}} e^{-19q} \leq \frac{512q}{59\sqrt{\cos 1}} e^{-19q} =: \kappa_F. \quad (\text{BRD37})$$

Every root-size threshold and every numerator degree has thus been included in the tail estimate.

6 The complete high relation determinants and signed return

Combining the actual positive-form inequalities BRD30 and BRD37 with BRD20 proves

$$0 \leq t_{a,r} \leq r \log(1 + \kappa_{I,r} + \kappa_F), \quad a = 0, f, \quad r = q, q + 1. \quad (\text{BRD38})$$

Indeed the eigenvalues of $D_{a,r}$ lie in $[0, \kappa_{I,r} + \kappa_F]$ by congruence and addition. The tails are the exact forms in BRD14 and BRD20; BRD38 is a proved estimate for their determinants, not an assumption that those forms are absent. In fact the difference of their two nonnegative logarithms has the sharper bound

$$|t_{f,r} - t_{0,r}| \leq r \log(1 + \kappa_{I,r} + \kappa_F). \quad (\text{BRD39})$$

Both terms belong to that same interval by BRD38.

For all original degrees satisfying BRD22, put

$$\mathcal{E}_r = 2r\Delta_\chi(\epsilon) + r\Delta_f(\epsilon) + r\log(1 + \kappa_{I,r} + \kappa_F). \quad (\text{BRD40})$$

Equations BRD18, BRD21 and BRD38 first give the stronger one-sided statement. If $U_r = 2lr\log q + \log(\det K_{l,r}/\det K_{0,r})$ and $L_r = r\log(1 + \kappa_{I,r} + \kappa_F)$, then

$$-2r\Delta_\chi(\epsilon) - L_r \leq \log \mathfrak{R}_{q+r-1} - U_r \leq 2r\Delta_\chi(\epsilon) + r\Delta_f(\epsilon) + L_r. \quad (\text{BRD41a})$$

In particular they prove the full, finite absolute bound

$$\left| \log \mathfrak{R}_{q+r-1} - 2lr\log q - \log \frac{\det K_{l,r}}{\det K_{0,r}} \right| \leq \mathcal{E}_r, \quad r = q, q+1. \quad (\text{BRD41})$$

For fixed original d, g, m , $\mathcal{E}_r = o(q)$. BRD19 proves the bulk part. The tail part is at most $r(\kappa_{I,r} + \kappa_F)$, which tends to zero exponentially by BRD30 and BRD37, with $r \leq q+1$. More precisely the bulk error is $O(k)$ for $m = 1$ and $O(1)$ for each fixed $m \geq 2$, while these displayed finite formulas give the actual constants and thresholds.

To return this calculation to the existing signed functional, retain its exact two-low-endpoint constant

$$C_{\text{rel}} = \log D_{l,q-1} + \log D_{l,q} - \log \mathfrak{r}_\chi - \log D_{l,2q-1} - \log D_{l,2q}, \quad D_{l,N} = \prod_{a=0}^N \prod_{j=0}^{2l-1} (a + j + 1/2).$$

Here $\mathfrak{r}_\chi$ is the unchanged original moment ratio $\int |f\chi|^2 d\sigma / \int |\chi|^2 d\sigma$. The previously proved source/relation determinant identity is $\mathfrak{T}_k = C_{\text{rel}} + \log \mathfrak{R}_{2q-1} + \log \mathfrak{R}_{2q}$; it retains the same full relation Grams as BRD10. Substitution of BRD41 therefore gives

$$\mathfrak{T}_k = C_{\text{rel}} + 2l(2q+1)\log q + \sum_{r=q}^{q+1} \log \frac{\det K_{l,r}}{\det K_{0,r}} + \varepsilon_k, \quad (\text{BRD42})$$

$$|\varepsilon_k| \leq \mathcal{E}_q + \mathcal{E}_{q+1} = o(q).$$

The incoming return $R_k^0 - R_k^\sigma$ is $-\mathfrak{T}_k$, so its entire displayed expression changes sign. The remaining objects are the two explicitly specified integral determinant ratios in BRD15 at the actual ranks q and $q+1$. Their evaluation, with the original Gamma density $\sigma(qz)$ and the retained deformation $\exp(a_k/z^2)$, is the next analytic calculation. This result closes the source truncation error at the $o(q)$ scale without assigning an unevaluated value to those integral determinants or to the independent arithmetic defect.

Independent exact exterior constants

7 Exact exterior constants for the original arithmetic polynomial

This calculation retains the original monic polynomial, its translated coordinate, and every root multiplicity. Let

$$\ell \in \mathbb{Z}_{\geq 0}, \quad k = 4\ell + 1, \quad m \in \mathbb{Z}_{\geq 1}, \quad e = 1 + k(m-1), \quad q = e(k+1)^2, \quad c \in \mathbb{C}, \quad \delta, \gamma \in \mathbb{R}. \quad (\text{IBC1})$$

Put $d = \delta^2$, $g = \gamma^2$, $K = k(k+2)$, and

$$n_a = 2a - k, \quad z_{ab} = n_a \delta + i n_b \gamma, \quad \chi(c+x) = \prod_{a,b=0}^k (x - z_{ab})^e. \quad (\text{IBC2})$$

In particular the degree is exactly q . Define the real logarithm of the following positive quotient on the original coordinate:

$$L_\chi(y) = \log \frac{|\chi(c+iy)|^2}{|y|^{2q}}, \quad y \in \mathbb{R} \setminus \{0\}, \quad \chi(c+iy) \neq 0. \quad (\text{IBC3})$$

All assertions about its exterior series assume

$$\tau = \frac{k^2(d+g)}{y^2} < 1. \quad (\text{IBC4})$$

This inequality makes each factor in (IBC2) nonzero at $x = iy$. The case $d = g = 0$ gives $\chi(c+x) = x^q$ and $L_\chi(y) = 0$; formulas with a quotient by a power of τ below use their displayed continuous value at $\tau = 0$.

7.1 The finite grid moments

Define the actual finite averages

$$\mu_{2r} = \frac{1}{k+1} \sum_{a=0}^k n_a^{2r}. \quad (\text{IBC5})$$

Odd moments vanish under the exact permutation $a \mapsto k-a$. The even moments needed here are

$$\mu_2 = \frac{K}{3}, \quad \mu_4 = \frac{K(3K-4)}{15}, \quad \mu_6 = \frac{K(3K^2-12K+16)}{21}. \quad (\text{IBC6})$$

Here is a derivation including the constants. Summing a finite geometric series gives, first for real $t \neq 0$ and then by continuity at zero,

$$\frac{1}{k+1} \sum_{a=0}^k e^{n_a t} = \frac{\sinh((k+1)t)}{(k+1) \sinh t}. \quad (\text{IBC7})$$

Multiply its Taylor series by $\sinh t$, and compare the coefficients of t^3, t^5, t^7 . Because $\mu_0 = 1$, the three resulting equations are

$$\begin{aligned} 3\mu_2 + 1 &= (k+1)^2, \\ 5\mu_4 + 10\mu_2 + 1 &= (k+1)^4, \\ 7\mu_6 + 35\mu_4 + 21\mu_2 + 1 &= (k+1)^6. \end{aligned} \quad (\text{IBC8})$$

Substitute $(k+1)^2 = K+1$ and solve successively. The numerators of μ_4 and μ_6 , before division by 5 and 7, respectively, are $K^2 - 4K/3$ and $K^3 - 4K^2 + 16K/3$. This proves (IBC6).

7.2 Convergent exterior series and its first three coefficients

For every y satisfying (IBC4) one has the absolutely convergent identity

$$L_\chi(y) = \sum_{r=1}^{\infty} \frac{p_{2r}}{y^{2r}}, \quad p_{2r} = \frac{e(-1)^{r+1}}{r} \operatorname{Re} \sum_{a,b=0}^k z_{ab}^{2r}. \quad (\text{IBC9})$$

To prove it, use the exact factorization

$$\frac{|\chi(c + iy)|^2}{|y|^{2q}} = \prod_{a,b=0}^k |1 + iz_{ab}/y|^{2e}. \quad (\text{IBC10})$$

The disk $|z_{ab}/y| < 1$ lies in the domain of the Taylor logarithm of $1 + iz_{ab}/y$. Thus the logarithm in (IBC3) equals $2e \operatorname{Re} \sum_{a,b} \log(1 + iz_{ab}/y)$. In the series $\log(1 + w) = \sum_{n \geq 1} (-1)^{n+1} w^n / n$, odd powers cancel under $(a, b) \mapsto (k - a, k - b)$, which sends z_{ab} to $-z_{ab}$. The term $n = 2r$ is precisely (IBC9). Absolute convergence follows from the finite root set and (IBC4); all rearrangements made here are therefore valid.

The first two coefficients are

$$p_2 = \frac{qk(k+2)}{3}(d-g) \quad (\text{IBC11})$$

and

$$\begin{aligned} p_4 &= q \left(3\mu_2^2 dg - \frac{\mu_4}{2}(d^2 + g^2) \right) \\ &= q \left[\frac{k^2(k+2)^2}{3} dg - \frac{k(k+2)(3k^2 + 6k - 4)}{30}(d^2 + g^2) \right]. \end{aligned} \quad (\text{IBC12})$$

Indeed,

$$\begin{aligned} \operatorname{Re} \sum_{a,b} z_{ab}^2 &= (k+1)^2 \mu_2(d-g), \\ \operatorname{Re} \sum_{a,b} z_{ab}^4 &= (k+1)^2 \{ \mu_4(d^2 + g^2) - 6\mu_2^2 dg \}. \end{aligned} \quad (\text{IBC13})$$

These identities follow by expanding each power and using the independent finite sums over a and b . Multiplication by e , and by $-e/2$ in the second line, proves (IBC11)–(IBC12). For use when an additional signed term is useful, the next coefficient is

$$\begin{aligned} p_6 &= \frac{q}{3} \{ \mu_6(d^3 - g^3) - 15\mu_4\mu_2 d^2 g + 15\mu_4\mu_2 dg^2 \} \\ &= \frac{q(d-g)}{3} \{ \mu_6(d^2 + dg + g^2) - 15\mu_4\mu_2 dg \}. \end{aligned} \quad (\text{IBC14})$$

The real terms of $(u + iv)^6 = u^6 - 15u^4v^2 + 15u^2v^4 - v^6 + i(6u^5v - 20u^3v^3 + 6uv^5)$ prove this formula after substituting $u = n_a\delta$, $v = n_b\gamma$.

7.3 Finite remainder estimates preserving the root radii

For $s \geq 1$ and $0 \leq t < 1$, set

$$h_s(t) = -\log(1-t) - \sum_{r=1}^{s-1} \frac{t^r}{r} = \sum_{r=s}^{\infty} \frac{t^r}{r}, \quad F_s(t) = \sum_{j=0}^{\infty} \frac{t^j}{s+j}. \quad (\text{IBC15})$$

Then $h_s(t) = t^s F_s(t)$, $F_s(0) = 1/s$, and F_s is nondecreasing because every coefficient of its power series is nonnegative. Define the following finite sums on the original roots:

$$\begin{aligned} r_{ab}(y) &= \frac{n_a^2 d + n_b^2 g}{y^2}, \\ \mathcal{B}_s(y) &= e \sum_{a,b=0}^k h_s(r_{ab}(y)), \\ M_{2s} &= e \sum_{a,b=0}^k |z_{ab}|^{2s}. \end{aligned} \tag{IBC16}$$

For each $s \geq 1$, the exact series (IBC9) gives

$$\left| L_\chi(y) - \sum_{r=1}^{s-1} \frac{p_{2r}}{y^{2r}} \right| \leq \mathcal{B}_s(y) \leq \frac{M_{2s}}{|y|^{2s}} F_s(\tau) \leq \frac{M_{2s}}{s|y|^{2s}(1-\tau)} \leq \frac{q\tau^s}{s(1-\tau)}. \tag{IBC17}$$

For the first inequality, take the absolute value of each term of the absolutely convergent tail, use $|\operatorname{Re} z_{ab}^{2r}| \leq |z_{ab}|^{2r}$, and sum $\sum_{r \geq s} r^r_{ab}/r$. For the second, use $r_{ab} \leq \tau$ and the monotonicity of F_s . For the third, each coefficient $1/(s+j)$ is at most $1/s$. For the last, $|z_{ab}|^2 \leq k^2(d+g)$ and there are exactly $e(k+1)^2 = q$ roots counted with multiplicity. This proves each comparison in (IBC17); the first bound is a finite sum and retains the individual radii instead of replacing them by their maximum.

In particular the remainder after the quadratic coefficient obeys the claimed estimate with exactly the stated factor:

$$L_\chi(y) = \frac{p_2}{y^2} + E_2(y), \quad |E_2(y)| \leq \frac{q\tau^2}{2(1-\tau)}. \tag{IBC18}$$

Two sharper finite estimates, available from the same calculation, are

$$|E_2(y)| \leq \mathcal{B}_2(y) \leq \frac{M_4}{y^4} F_2(\tau) \leq q\{-\log(1-\tau) - \tau\}, \tag{IBC19}$$

where

$$M_4 = q\{\mu_4(d^2 + g^2) + 2\mu_2^2 dg\}. \tag{IBC20}$$

To verify the last step of (IBC19), use $M_4/y^4 \leq q\tau^2$ and $\tau^2 F_2(\tau) = h_2(\tau)$. Expanding $(n_a^2 d + n_b^2 g)^2$ proves (IBC20).

Retaining the signed fourth-order term yields

$$L_\chi(y) = \frac{p_2}{y^2} + \frac{p_4}{y^4} + E_3(y), \quad |E_3(y)| \leq \mathcal{B}_3(y) \leq \frac{M_6}{|y|^6} F_3(\tau) \leq \frac{M_6}{3|y|^6(1-\tau)}, \tag{IBC21}$$

with the exact moment

$$M_6 = q\{\mu_6(d^3 + g^3) + 3\mu_4\mu_2 dg(d+g)\}. \tag{IBC22}$$

This time expand $(n_a^2 d + n_b^2 g)^3$, and use (IBC6). Consequently a signed enclosure for E_2 , which can be intersected with (IBC19), is

$$\frac{p_4}{y^4} - \mathcal{B}_3(y) \leq E_2(y) \leq \frac{p_4}{y^4} + \mathcal{B}_3(y). \tag{IBC23}$$

If (IBC14) is also retained, (IBC17) with $s = 4$ gives the finite error $\mathcal{B}_4(y) \leq q\tau^4/[4(1-\tau)]$. In the case $d = g$, the computed coefficients explicitly read

$$p_2 = p_6 = 0, \quad p_4 = \frac{2qK(K+2)}{15} d^2. \tag{IBC24}$$

This is a consequence of (IBC11)–(IBC14), not the removal of any term from the original polynomial.

7.4 Exact second and fourth powers of the Gamma-factor offsets

Retain the monic polynomial

$$f_\ell(S) = \prod_{j=0}^{\ell-1} (S - c - t_j), \quad t_j = 2j + \frac{1}{2}. \quad (\text{IBC25})$$

For $\ell = 0$ it is the constant polynomial 1. Its two needed power sums are

$$\begin{aligned} T_2 &:= \sum_{j=0}^{\ell-1} t_j^2 = \frac{\ell(16\ell^2 - 12\ell - 1)}{12} \\ &= \frac{(k-1)(k^2 - 5k + 3)}{48}, \end{aligned} \quad (\text{IBC26})$$

and

$$\begin{aligned} T_4 &:= \sum_{j=0}^{\ell-1} t_j^4 = \frac{\ell(768\ell^4 - 960\ell^3 - 160\ell^2 + 360\ell + 7)}{240} \\ &= \frac{(k-1)(3k^4 - 27k^3 + 53k^2 + 53k - 75)}{960}. \end{aligned} \quad (\text{IBC27})$$

For completeness put $S_r(n) = \sum_{j=0}^{n-1} j^r$. The exact formulas

$$\begin{aligned} S_0(n) &= n, & S_1(n) &= \frac{n(n-1)}{2}, \\ S_2(n) &= \frac{n(n-1)(2n-1)}{6}, & S_3(n) &= \frac{n^2(n-1)^2}{4}, \\ S_4(n) &= \frac{n(n-1)(2n-1)(3n^2 - 3n - 1)}{30} \end{aligned} \quad (\text{IBC28})$$

are proved by finite differences: each right side vanishes at $n = 0$, and its value at $n + 1$ minus its value at n , after expansion, is n^r for its respective r . Expanding the original offsets gives

$$\begin{aligned} T_2 &= 4S_2(\ell) + 2S_1(\ell) + \ell/4, \\ T_4 &= 16S_4(\ell) + 16S_3(\ell) + 6S_2(\ell) + S_1(\ell) + \ell/16. \end{aligned} \quad (\text{IBC29})$$

Substitute (IBC28), collect the displayed powers of ℓ , and then substitute $\ell = (k-1)/4$. This proves both forms of (IBC26)–(IBC27), including their zero values for the empty sums.

These constants control the actual Gamma polynomial on the same line without an additional exterior restriction:

$$L_f(y) := \log \frac{|f_\ell(c + iy)|^2}{|y|^{2\ell}} = \sum_{j=0}^{\ell-1} \log(1 + t_j^2/y^2), \quad y \in \mathbb{R} \setminus \{0\}. \quad (\text{IBC30})$$

For all such y ,

$$0 \leq \frac{T_2}{y^2} - L_f(y) \leq \frac{T_4}{2y^4}. \quad (\text{IBC31})$$

After retaining the signed fourth-order coefficient one has the stronger one-sided statement

$$\begin{aligned} L_f(y) &= \frac{T_2}{y^2} - \frac{T_4}{2y^4} + R_f(y), \\ \sum_{j=0}^{\ell-1} \frac{t_j^6}{3|y|^6(1 + t_j^2/y^2)} &\leq R_f(y) \leq \sum_{j=0}^{\ell-1} \frac{t_j^6}{3|y|^6}. \end{aligned} \quad (\text{IBC32})$$

Indeed, for every real $x \geq 0$, elementary integration gives

$$x - \log(1+x) = \int_0^x \frac{t}{1+t} dt, \quad \log(1+x) - x + \frac{x^2}{2} = \int_0^x \frac{t^2}{1+t} dt. \quad (\text{IBC33})$$

The first integral lies between 0 and $x^2/2$. The second lies between $x^3/[3(1+x)]$ and $x^3/3$, because $1/(1+x) \leq 1/(1+t) \leq 1$ on the entire integration interval. Apply these inequalities to each $x = t_j^2/y^2$ in (IBC30). This proves (IBC31)–(IBC32) and keeps all offset factors.

Independent control of both discarded intervals

8 Independent original-source control of both discarded intervals

This proof uses the original relation polynomial, the original Gamma density, and the original polynomial coefficient spaces. The change of variable below is an explicit evaluation map, with its complete coefficient matrix. No mass factor or polynomial degree is removed from a Gram matrix.

8.1 Objects, intervals, and the coefficient map

Fix integers $m \geq 1$ and $l \geq 1$, and put

$$k = 4l + 1, \quad c = k/2, \quad e = 1 + k(m - 1), \quad q = e(k + 1)^2, \quad 0 < \delta < \frac{1}{2}, \quad \gamma > 2. \quad (\text{FTC1})$$

Write $d = \delta^2$ and $g_\gamma = \gamma^2$. The notation g_γ does not reassign the arithmetic entire function denoted by g elsewhere. Define

$$\begin{aligned} w_{ab} &= (2a - k)\delta + i(2b - k)\gamma, & 0 \leq a, b \leq k, \\ \chi(S) &= \prod_{a,b=0}^k (S - c - w_{ab})^e, & R_{\max} = k\sqrt{d + g_\gamma}, \\ t_j &= 2j + \frac{1}{2}, \quad f_l(S) = \prod_{j=0}^{l-1} (S - c - t_j), & t_{\max} = 2l - \frac{3}{2}, \\ \sigma(y) &= \frac{|\Gamma(\frac{1}{4} + iy/2)|^2}{2\pi}, & \phi_0(S) = 1, \quad \phi_l(S) = f_l(S). \end{aligned} \quad (\text{FTC2})$$

Thus $|w_{ab}| \leq R_{\max}$, the multiset of roots w_{ab} is invariant under $w \mapsto -w$, and $|\phi_l(c + iy)|^2 = \prod_{j=0}^{l-1} (y^2 + t_j^2)$. The two relation dimensions treated here are $r = q$ and $r = q + 1$. For $P(S) = \sum_{n=0}^{r-1} p_n S^n$ and a Borel set $A \subset \mathbb{R}$ define the original form

$$\mathcal{H}_{s,A}(P, P) = \int_{\{y: y/q \in A\}} |P(c + iy)\chi(c + iy)\phi_s(c + iy)|^2 \sigma(y) dy, \quad s \in \{0, l\}. \quad (\text{FTC3})$$

Its matrix $H_{s,A}$ is in the unchanged ordered basis $(1, S, \dots, S^{r-1})$; in particular its (i, j) entry, indexed from zero, contains $\overline{(c + iy)^i} (c + iy)^j$. Set

$$Q(z) = P(c + iqz), \quad (T_{c,q})_{jn} = \begin{cases} \binom{n}{j} c^{n-j} (iq)^j, & j \leq n, \\ 0, & j > n. \end{cases} \quad \det T_{c,q} = (iq)^{r(r-1)/2}. \quad (\text{FTC4})$$

The coefficient vector of Q is exactly $T_{c,q}p$. If $\tilde{H}_{s,A}$ is the matrix in powers of z for the measure obtained by the literal substitution $y = qz$, including the Jacobian q , then $H_{s,A} = T_{c,q}^* \tilde{H}_{s,A} T_{c,q}$ and $\det H_{s,A} = q^{r(r-1)} \det \tilde{H}_{s,A}$. All subsequent comparisons are comparisons of the original matrices $H_{s,A}$. These congruences prove their exact relation to the intermediate z coordinate calculation.

8.2 Gamma bounds with the original mass

For $0 < \theta < \pi/2$ and every real y ,

$$\frac{\pi}{\Gamma(3/4)^2} e^{-\pi|y|} \leq \sigma(y) \leq \frac{\Gamma(1/4)^2}{2\pi\sqrt{\cos\theta}} e^{-\theta|y|}. \quad (\text{FTC5})$$

Here is a proof of the constants. For $a > 0$ and $b \geq 0$, rotate the Euler integral for $\Gamma(a + ib)$ through the angle θ in the first quadrant, using the branch of the logarithm with argument in $[0, \theta]$. The small circular arc is $O(\varepsilon^a)$, and the large arc tends to zero because $\cos v \geq \cos \theta > 0$ on the rotation sector. Consequently

$$\Gamma(a + ib) = e^{i\theta(a+ib)} \int_0^\infty t^{a+ib-1} e^{-e^{i\theta}t} dt, \quad |\Gamma(a + ib)| \leq e^{-\theta b} (\cos \theta)^{-a} \Gamma(a).$$

For $b < 0$ use the conjugate sector. Squaring with $a = 1/4$, $b = y/2$ proves the upper bound in (FTC5).

For completeness, the reflection identity used for the lower bound follows from Euler's double integral, with the substitution $u = vt$ followed by integration in v :

$$\Gamma(z)\Gamma(1-z) = \int_0^\infty \frac{t^{z-1}}{1+t} dt, \quad 0 < \operatorname{Re} z < 1.$$

On a keyhole contour with $0 < \arg t < 2\pi$, the integrals over the small and large circles vanish because $0 < \operatorname{Re} z < 1$. The two straight integrals have difference $(1 - e^{2\pi iz})$ times the displayed integral, whereas the residue at -1 is $-e^{\pi iz}$. The residue formula therefore gives $\Gamma(z)\Gamma(1-z) = \pi/\sin(\pi z)$. The unrotated Euler integral gives $|\Gamma(3/4 - iy/2)| \leq \Gamma(3/4)$, and

$$|\sin(\pi/4 + i\pi y/2)|^2 = \frac{1}{2} \cosh(\pi y).$$

It follows that

$$\sigma(y) \geq \frac{\pi}{\Gamma(3/4)^2 \cosh(\pi y)} \geq \frac{\pi}{\Gamma(3/4)^2} e^{-\pi|y|}.$$

In particular the exact ratio of the upper and lower constant in (FTC5) is $(\cos \theta)^{-1/2}$, since the same reflection calculation at $z = 1/4$ gives $\Gamma(1/4)\Gamma(3/4) = \pi\sqrt{2}$. Denote the lower constant by $c_L = \pi/\Gamma(3/4)^2$.

8.3 Exterior expansion of the unchanged relation polynomial

Define

$$a_k = \frac{k(k+2)(d-g_\gamma)}{3q} < 0, \quad \tau_x = \frac{k^2(d+g_\gamma)}{q^2x^2}, \quad \Delta_\chi(x) = \frac{q\tau_x^2}{2(1-\tau_x)} \quad (\tau_x < 1). \quad (\text{FTC6})$$

For real $|z| \geq x$ with $\tau_x < 1$ the exact logarithmic expansion is

$$\log \frac{|\chi(c + iqz)|^2}{(q|z|)^{2q}} = \frac{a_k}{z^2} + E_k(z), \quad |E_k(z)| \leq \Delta_\chi(x). \quad (\text{FTC7})$$

Indeed the series for $\log(1 - w_{ab}/(iqz))$ converges absolutely and uniformly on this real exterior set. Pairing w and $-w$ cancels every odd power. Its degree-two term is

$$\frac{e}{q^2z^2} \sum_{a,b=0}^k \operatorname{Re}(w_{ab}^2) = \frac{e(k+1)^2k(k+2)(d-g_\gamma)}{3q^2z^2} = \frac{a_k}{z^2},$$

where $\sum_{a=0}^k (2a-k)^2 = k(k+1)(k+2)/3$ follows by expanding the square and summing a and a^2 . The remaining series is

$$E_k(z) = -e \operatorname{Re} \sum_{a,b=0}^k \sum_{n=2}^\infty \frac{1}{n} \left(\frac{w_{ab}}{iqz} \right)^{2n}.$$

Thus its absolute value is at most $q \sum_{n=2}^\infty \tau_x^n / n \leq q\tau_x^2/[2(1-\tau_x)]$. Also put

$$\Delta_{f,0}(x) = 0, \quad \Delta_{f,l}(x) = \frac{\sum_{j=0}^{l-1} t_j^2}{q^2x^2}. \quad (q|z|)^{2s} \leq |\phi_s(c + iqz)|^2 \leq (q|z|)^{2s} e^{\Delta_{f,s}(x)} \quad (|z| \geq x). \quad (\text{FTC8})$$

The last upper estimate is exactly $\log(1+u) \leq u$ applied separately to every factor; none of the t_j have been changed.

8.4 A polynomial estimate on the original coefficient space

For every complex polynomial Q of degree at most $r - 1$, write $I_Q = \int_1^2 |Q(t)|^2 dt$. For real $|z| \geq B \geq 2$,

$$|Q(z)|^2 \leq r^2 ((4 + 6/B)|z|)^{2(r-1)} I_Q. \quad (\text{FTC9})$$

For $0 < \varepsilon \leq 1/2$ and real $|z| \leq \varepsilon$,

$$|Q(z)|^2 \leq r^2 (6 + 4\varepsilon)^{2(r-1)} I_Q. \quad (\text{FTC10})$$

To prove both statements, use the polynomials $\psi_n(t) = \sqrt{2n+1} P_n(2t-3)$, where

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n.$$

Integration by parts proves orthogonality to every lower-degree polynomial: all boundary derivatives of $(x^2 - 1)^n$ of order less than n vanish. The leading coefficient is $(2n)!/[2^n(n!)^2]$, and the same integration by parts, together with

$$\int_{-1}^1 (1 - x^2)^n dx = \frac{2^{2n+1} (n!)^2}{(2n+1)!},$$

gives $\int_{-1}^1 P_n(x)^2 dx = 2/(2n+1)$. The last integral follows from $x = 2t - 1$ and n integrations by parts for $\int_0^1 t^n (1-t)^n dt = (n!)^2/(2n+1)!$. Hence $(\psi_n)_{0 \leq n < r}$ is an orthonormal basis on $[1, 2]$.

For real $x \geq 1$ the finite integral formula

$$P_n(x) = \frac{1}{\pi} \int_0^\pi (x + \sqrt{x^2 - 1} \cos v)^n dv$$

is obtained by expanding the integrand: odd cosine moments vanish and the $2j$ -th moment is $\binom{2j}{j}/4^j$, giving the same polynomial as the displayed Rodrigues derivative. It implies $|P_n(x)| \leq (2x)^n$. Parity of Rodrigues' derivative gives this bound with $2|x|$ for $x \leq -1$ as well. Expansion of Q in the orthonormal basis and complex Cauchy–Schwarz now give

$$|Q(z)|^2 \leq I_Q \sum_{n=0}^{r-1} (2n+1) |P_n(2z-3)|^2.$$

On the exterior set, $|2z-3| \geq 1$ and $2|2z-3| \leq 4|z| + 6 \leq (4 + 6/B)|z|$. On the inner interval, $|2z-3| \geq 2$ and $2|2z-3| \leq 6 + 4\varepsilon$. Each bound on the right is at least one, and $\sum_{n=0}^{r-1} (2n+1) = r^2$. This proves (FTC9)–(FTC10), including their degree and constants.

8.5 Far interval: a finite bound and a uniform exponential bound

Let $\mathcal{B}_B = \{z : 1 \leq |z| \leq B\}$ and $\mathcal{F}_B = \{z : |z| > B\}$, where $B \geq 2$. Suppose first that $\tau_1 < 1$, and set $\eta = -a_k + \Delta_\chi(1)$. The positive segment $[1, 2]$ gives

$$\mathcal{H}_{s, \mathcal{B}_B}(P, P) \geq q^{2q+2s+1} c_L e^{-2\pi q - \eta} I_Q. \quad (\text{FTC11})$$

In fact $a_k/z^2 \geq a_k$ on this segment, the factor z^{2q+2s} is at least one, and (FTC5), (FTC7), (FTC8), and the Jacobian $dy = q dz$ give exactly this inequality.

Write $D_s = 2q + 2s + 2r - 2$, so that $D_s \geq 0$. If $0 < \theta < \pi/2$ and $\theta q > D_s/B$, then

$$\begin{aligned} \mathcal{H}_{s, \mathcal{F}_B} &\leq \kappa_{s, r}^{\text{far}}(B, \theta) \mathcal{H}_{s, \mathcal{B}_B}, \\ \kappa_{s, r}^{\text{far}}(B, \theta) &= \frac{2r^2}{\sqrt{\cos \theta}} e^{2\pi q + \eta + \Delta_\chi(B) + \Delta_{f, s}(B)} (4 + 6/B)^{2(r-1)} \frac{B^{D_s} e^{-\theta q B}}{\theta q - D_s/B}. \end{aligned} \quad (\text{FTC12})$$

These are quadratic-form inequalities on the full original relation space. To verify the numerator, use $a_k/z^2 \leq 0$ in the far interval, then (FTC5), (FTC7)–(FTC9). The integral over its two components is bounded by

$$\frac{2\Gamma(1/4)^2}{2\pi\sqrt{\cos\theta}} q^{2q+2s+1} r^2 (4+6/B)^{2(r-1)} e^{\Delta_\chi(B)+\Delta_{f,s}(B)} I_Q \int_B^\infty z^{D_s} e^{-\theta q z} dz.$$

For $z \geq B$, $D_s \log(z/B) \leq D_s(z-B)/B$ because $D_s \geq 0$. Integration of the resulting exponential proves

$$\int_B^\infty z^{D_s} e^{-\theta q z} dz \leq \frac{B^{D_s} e^{-\theta q B}}{\theta q - D_s/B}.$$

Divide by (FTC11) and use the exact Gamma-constant ratio proved after (FTC5). This proves (FTC12).

For $B = 64$, $\theta = 1$, the original degrees give $D_s \leq 4q + 2l \leq 5q$, since $r - 1 \leq q$ and $2l \leq q$. Therefore

$$\begin{aligned} \kappa_{s,r}^{\text{far}}(64, 1) &\leq \frac{128r^2}{59q\sqrt{\cos 1}} e^{\eta + \Delta_\chi(64) + \Delta_{f,s}(64) - c_* q}, \\ c_* &= 64 - 2\pi - 5 \log 64 - 2 \log(131/32) > 20. \end{aligned} \quad (\text{FTC13})$$

Here $4 + 6/64 = 131/32$, $q - D_s/64 \geq 59q/64$, and $r - 1 \leq q$. The strict numerical bound uses $\pi < 4$, $\log 2 < 1$, and $131/32 < 8$, which give $2\pi + 5 \log 64 + 2 \log(131/32) < 8 + 30 + 6 = 44$.

There is an explicit finite domain on which this becomes uniform in the original packet parameters:

$$q \geq 2k\sqrt{d+g_\gamma} \implies \kappa_{s,r}^{\text{far}}(64, 1) \leq \frac{512q}{59\sqrt{\cos 1}} e^{-19q}. \quad (\text{FTC14})$$

Indeed $\tau_1 \leq 1/4$ and $k \geq 5$ give, separately,

$$-a_k \leq \frac{k+2}{3k} q \tau_1 \leq \frac{7q}{60}, \quad \Delta_\chi(1) \leq \frac{q}{24}, \quad \Delta_\chi(64) \leq \frac{q}{24}.$$

Their sum is at most $q/5$. Since $t_j < 2l$, $\sum_j t_j^2 \leq 4l^3$ and $2l \leq q$ imply $\Delta_{f,l}(64) \leq q/8192$. Thus the sum of all three exponent corrections in (FTC13) is strictly less than $q/4$. Equation (FTC13), $c_* > 20$, and $r \leq q + 1 \leq 2q$ prove (FTC14). The threshold is a stated domain for a finite explicit bound; for each fixed original δ, γ, m it holds for all sufficiently large $k = 4l + 1$, because $q \geq k^2$.

8.6 Inner interval: the exact original-root product bound

Set $\varepsilon = 2^{-10}$, $\mathcal{I}_\varepsilon = \{z : |z| < \varepsilon\}$, and suppose

$$R_{\max}/q \leq \varepsilon, \quad t_{\max}/q \leq \varepsilon. \quad (\text{FTC15})$$

The second inequality follows from the first for these original parameters: $t_{\max} < k/2$ whereas $R_{\max} > 2k$. Pair the q roots of $\chi(c+S)$, with their multiplicities, as $w, -w$. There are exactly $q/2$ pairs. Thus

$$\begin{aligned} |\chi(c+iy)|^2 &= \prod_{\{w, -w\}} |y^2 + w^2|^2 \leq (y^2 + R_{\max}^2)^q, \\ |\chi(c+iqz)|^2 &\leq q^{2q} (2\varepsilon^2)^q \quad (|z| \leq \varepsilon). \end{aligned} \quad (\text{FTC16})$$

On the segment $1 \leq z \leq 2$ the same paired product, with the reverse triangle inequality, gives

$$|\chi(c+iqz)|^2 \geq q^{2q} (1 - R_{\max}^2/q^2)^q \geq q^{2q} (3/4)^q. \quad (\text{FTC17})$$

For the numerator factor, the original roots give

$$|\phi_s(c+iqz)|^2 \leq q^{2s} (2\varepsilon^2)^s \quad (|z| \leq \varepsilon), \quad |\phi_s(c+iqz)|^2 \geq q^{2s} \quad (1 \leq z \leq 2). \quad (\text{FTC18})$$

The factor $(2\varepsilon^2)^s$ is kept in the following stronger bound:

$$\begin{aligned}\mathcal{H}_{s,\mathcal{I}_\varepsilon} &\leq \kappa_{s,r}^{\text{in}} \mathcal{H}_{s,\mathcal{B}_{64}}, \\ \kappa_{s,r}^{\text{in}} &\leq \frac{2\varepsilon r^2}{\sqrt{\cos 1}} (2\varepsilon^2)^s \beta^q \leq \frac{2\varepsilon r^2}{\sqrt{\cos 1}} e^{-2q}, \quad \beta = \frac{4}{3} 2^{-13} e^{2\pi} < e^{-2}.\end{aligned}\tag{FTC19}$$

To prove the first inequality, bound $\sigma(qz)$ above by $\Gamma(1/4)^2/(2\pi\sqrt{\cos 1})$ on the inner interval, using (FTC5) and $e^{-|qz|} \leq 1$. Equations (FTC10) and (FTC16)–(FTC18) bound its integral by

$$2\varepsilon q^{2q+2s+1} \frac{\Gamma(1/4)^2}{2\pi\sqrt{\cos 1}} r^2 (6+4\varepsilon)^{2(r-1)} (2\varepsilon^2)^{q+s} I_Q.$$

The bulk denominator from [1, 2] is at least $q^{2q+2s+1} c_L (3/4)^q e^{-2\pi q} I_Q$. The ratio of Gamma constants is again exactly $(\cos 1)^{-1/2}$. Since $6+4\varepsilon < 8$ and $r-1 \leq q$, the polynomial factor is at most $8^{2q} = 2^{6q}$, and $2\varepsilon^2 \cdot 2^6 = 2^{-13}$. This proves the displayed value of β and the first bound in (FTC19).

The elementary strict estimate on β can be certified without decimal approximations. One has $\log(4/3) < 1/3$ from $\log(1+x) < x$, and $\log 2 = 2 \sum_{n \geq 0} 3^{-2n-1}/(2n+1) > 2/3$, obtained by integrating the geometric series for $(1-t^2)^{-1}$ from zero to $1/3$. Polynomial division gives

$$\frac{x^4(1-x)^4}{1+x^2} = x^6 - 4x^5 + 5x^4 - 4x^2 + 4 - \frac{4}{1+x^2}.$$

Integrating on $[0, 1]$ proves $22/7 - \pi = \int_0^1 x^4(1-x)^4/(1+x^2)dx > 0$. It follows that

$$\log \beta < \frac{1}{3} - \frac{26}{3} + \frac{44}{7} = -\frac{43}{21} < -2.$$

Finally $(2\varepsilon^2)^s \leq 1$, proving the last bound in (FTC19).

8.7 Exact determinant transfer after both intervals are included

Define the complete central set

$$\mathcal{C} = \{z : \varepsilon \leq |z| \leq 64\}.\tag{FTC20}$$

It includes the interval $\varepsilon < |z| < 1$. Except for endpoints of Lebesgue measure zero, the sets $\mathcal{I}_\varepsilon$, $\mathcal{C}$, and $\mathcal{F}_{64}$ partition the entire original real integration line after $y = qz$. All four matrices in

$$H_{s,\mathbb{R}} = H_{s,\mathcal{C}} + H_{s,\mathcal{I}_\varepsilon} + H_{s,\mathcal{F}_{64}}, \quad 0 \leq H_{s,\mathbb{R}} - H_{s,\mathcal{C}} \leq \kappa_{s,r} H_{s,\mathcal{C}}, \quad \kappa_{s,r} = \kappa_{s,r}^{\text{in}} + \kappa_{s,r}^{\text{far}}(64, 1)\tag{FTC21}$$

are in the original basis. The last inequality follows from (FTC12) and (FTC19), because $\mathcal{B}_{64} \subset \mathcal{C}$ and integrals of nonnegative functions preserve that inclusion. Thus neither the central annulus nor any cross term of a polynomial Gram matrix has been discarded.

The matrix $H_{s,\mathcal{C}}$ is positive definite: σ is positive, the fixed polynomials $\chi\phi_s$ have only finitely many real-line zeros, and a nonzero polynomial P cannot vanish on an interval. The Gamma upper bound proves convergence of every entry on the full line. Let L be the Cholesky factor defined by $H_{s,\mathcal{C}} = LL^*$. The explicit matrix

$$A = L^{-1}(H_{s,\mathbb{R}} - H_{s,\mathcal{C}})L^{-*}$$

has $0 \leq A \leq \kappa_{s,r} I$ by (FTC21), and $H_{s,\mathbb{R}} = L(I + A)L^*$. Hence, without changing the original determinants,

$$0 \leq \delta_{s,r} := \log \frac{\det H_{s,\mathbb{R}}}{\det H_{s,\mathcal{C}}} = \log \det(I + A) \leq r \log(1 + \kappa_{s,r}).\tag{FTC22}$$

This argument also proves the same far-only determinant bound if the inner interval has already been added: its base matrix then dominates $H_{s,\mathcal{B}_{64}}$.

At the original high endpoint $N = q + r - 1$, define, in precisely the same relation basis $(\chi, \chi S, \dots, \chi S^{r-1})$,

$$\mathcal{R}_N^{\text{all}} = \frac{\det H_{l,\mathbb{R}}}{\det H_{0,\mathbb{R}}}, \quad \mathcal{R}_N^{\mathcal{C}} = \frac{\det H_{l,\mathcal{C}}}{\det H_{0,\mathcal{C}}}.$$

Their exact signed relation is

$$\log \mathcal{R}_N^{\text{all}} - \log \mathcal{R}_N^{\mathcal{C}} = \delta_{l,r} - \delta_{0,r} \in [-r \log(1 + \kappa_{0,r}), r \log(1 + \kappa_{l,r})]. \quad (\text{FTC23})$$

If the numerator is written in the original Gamma-convolution density

$$r_k^0(y) = \frac{(\sqrt{2\pi})^k}{c_{k/4}} \frac{|\Gamma(k/4 + iy/2)|^2}{2\pi} = C_k |f_l(c + iy)|^2 \sigma(y),$$

$$c_{k/4} = 2^{1-k/2} \Gamma(k/2), \quad C_k = \frac{(\sqrt{2\pi})^k}{c_{k/4}} 4^{-l},$$

the displayed density identity follows by l applications of $\Gamma(z+1) = z\Gamma(z)$ at $z = 1/4 + iy/2$. The recurrence itself follows by integration by parts in Euler's integral, with both boundary terms zero for $\text{Re } z > 0$; squaring the l factors gives exactly $4^{-l} \prod_j (y^2 + t_j^2)$. Its Gram matrix is exactly $C_k H_{l,A}$, so its determinant is exactly $C_k^r \det H_{l,A}$. The term $r \log C_k$ is present both before and after the restriction and cancels in (FTC23); its original value is never set to one.

Under (FTC15), both tails have the common bound

$$\kappa_{s,r} \leq \mathcal{K}_q := \frac{1}{\sqrt{\cos 1}} \left(8\epsilon q^2 e^{-2q} + \frac{512q}{59} e^{-19q} \right). \quad (\text{FTC24})$$

Indeed (FTC15) implies the domain of (FTC14), and $r \leq 2q$ in (FTC19). In particular the sum of the two original high-endpoint changes satisfies

$$\left| \sum_{N \in \{2q-1, 2q\}} \left(\log \mathcal{R}_N^{\text{all}} - \log \mathcal{R}_N^{\mathcal{C}} \right) \right| \leq (2q+1) \log(1 + \mathcal{K}_q) \leq (2q+1) \mathcal{K}_q = o(q). \quad (\text{FTC25})$$

The two dimensions in this equation remain q and $q+1$, respectively. The last assertion follows directly from $q^j e^{-2q} \rightarrow 0$ for each fixed nonnegative integer j , as follows, for example, by retaining the $(j+1)$ -st positive term in the exponential power series.

Finally, the sufficient finite threshold for all of (FTC15)–(FTC25) is

$$q \geq 1024k\sqrt{d + g_\gamma}. \quad (\text{FTC26})$$

For fixed original parameters this follows from $k \geq 1024\sqrt{d + g_\gamma}$, since $q \geq k^2$. If a companion central-density Taylor expansion requires the stronger condition $\tau_\epsilon \leq 1/4$, the explicit sufficient threshold is $q \geq 2048k\sqrt{d + g_\gamma}$ instead. This latter threshold implies (FTC26); it is not tacitly inferred from equality in (FTC26), which only gives $\tau_\epsilon \leq 1$. All estimates above retain the original $\delta, \gamma, m, k, q, r$ and the original Gamma density throughout.

Exact contrast transport and the universal Gamma Hankel return

9 Rank- l transport of the original relation-Gram ratio

This calculation keeps the two fixed polynomial inclusions and transports their source form through the same positive density. The difference of their orthogonal projectors has precisely l positive and l negative eigenvalues. Its signed derivative controls the complete Gram-determinant ratio by the degree of the original multiplication polynomial.

9.1 Fixed coefficient columns, forms, and exact intersection

Let $0 < \epsilon < 64$, put $\mathcal{K}_\epsilon = [-64, -\epsilon] \cup [\epsilon, 64]$, and retain the evaluation coordinate $S = c + iQz$, where $c \in \mathbb{R}$, $Q > 0$. Use a finite positive Borel measure w on $\mathcal{K}_\epsilon$ whose support is infinite. This is a specified domain of the Gram construction; the concrete original densities used below are strictly positive on both intervals, and their finiteness and support are verified below. Let ψ be a bounded real measurable function, $w_t = e^{t\psi}w$ for $0 \leq t \leq 1$, and retain a monic polynomial

$$f(S) = \sum_{j=0}^l f_j S^j, \quad f_l = 1, \quad l \geq 1, \quad r \geq l, \quad n = r + l. \quad (\text{CTR1})$$

The abstract polynomial spaces and their coefficient columns are

$$\begin{aligned} \mathcal{H} &= \mathbb{C}[S]_{\leq r+l-1}, & \mathcal{P} &= \mathbb{C}[S]_{\leq r-1}, & \mathcal{F} &= f\mathcal{P}, \\ C_0 &= [e_0, \dots, e_{r-1}], & C_f &= [\text{coeff}(fS^j)]_{j=0}^{r-1}. \end{aligned} \quad (\text{CTR2})$$

Here e_j is the coefficient column of S^j in the original ordered ambient basis $1, S, \dots, S^{n-1}$. Thus C_0, C_f are fixed $n \times r$ matrices, and $(C_f)_{a,j} = f_{a-j}$, where a coefficient outside $0, \dots, l$ is zero. Multiplication by a nonzero polynomial is injective, so both matrices have rank r .

The original source form, its two restriction Grams, and the full ratio are

$$\begin{aligned} (M_t)_{a,b} &= \int_{\mathcal{K}_\epsilon} \overline{(c + iQz)^a} (c + iQz)^b e^{t\psi(z)} dw(z), \quad 0 \leq a, b < n, \\ G_0(t) &= C_0^* M_t C_0, & G_f(t) &= C_f^* M_t C_f, \\ \mathcal{R}_{w_t}(f) &= \frac{\det G_f(t)}{\det G_0(t)}. \end{aligned} \quad (\text{CTR3})$$

Every entry is finite because the coordinate and ψ are bounded on the compact set and w is finite. Every M_t is positive definite: a nonzero polynomial has finitely many zeros on the injective line $z \mapsto c + iQz$, whereas the support of w is infinite; at a support point outside those zeros it is bounded away from zero on a neighbourhood of positive measure. Multiplication by the positive bounded density $e^{t\psi}$ preserves that conclusion. Both restriction Grams are therefore positive definite.

There is an exact sequence, including all coefficient maps,

$$0 \longrightarrow \mathbb{C}[S]_{\leq r-l-1} \xrightarrow{a \mapsto (-fa, a)} \mathcal{P} \oplus \mathcal{P} \xrightarrow{(u,v) \mapsto u+fv} \mathcal{H} \longrightarrow 0. \quad (\text{CTR4})$$

The convention $\mathbb{C}[S]_{\leq -1} = 0$ applies at $r = l$. Indeed, division of $h \in \mathcal{H}$ by the original monic f gives $h = fv + u$ with $\deg v \leq r - 1$ and $\deg u < l \leq r$. The kernel is $u = -fv$, and its degree constraint forces $\deg v \leq r - l - 1$. This proves both surjectivity and the kernel identity. In particular,

$$\mathcal{I} := \mathcal{P} \cap \mathcal{F} = f\mathbb{C}[S]_{\leq r-l-1}, \quad \dim \mathcal{I} = r - l, \quad \mathcal{P} + \mathcal{F} = \mathcal{H}. \quad (\text{CTR5})$$

No source metric was needed for this exact polynomial intersection.

For any one of the fixed injective column matrices C define

$$P_C(t) = C(C^*M_tC)^{-1}C^*M_t. \quad (\text{CTR6})$$

Direct multiplication gives $P_C^2 = P_C$, $P_C C = C$ and $P_C^* M_t = M_t P_C$. Also $C^* M_t(x - P_C x) = 0$, proving that this is the actual M_t -orthogonal projector onto the column space of C . Cyclicity gives $\text{Tr } P_C = \text{Tr } I_{\text{rank } C}$. Let $P_0 = P_{C_0}$, $P_f = P_{C_f}$ and

$$D_t = P_f(t) - P_0(t), \quad D_t^* M_t = M_t D_t, \quad \text{Tr } D_t = r - r = 0. \quad (\text{CTR7})$$

9.2 The complete residual frame and its signed spectrum

The columns $C_I = [\text{coeff}(fS^j)]_{j=0}^{r-l-1}$ give the intersection, and P_I denotes CTR6 for C_I ; if $r = l$, $P_I = 0$. The inclusions $\mathcal{I} \subset \mathcal{P}, \mathcal{F}$ imply $P_0 P_I = P_f P_I = P_I$ and, by the adjoint equations, $P_I P_0 = P_I P_f = P_I$. Define the following literal columns:

$$\begin{aligned} U_0 &= (I - P_I)[e_0, \dots, e_{l-1}], \\ U_f &= (I - P_I)[\text{coeff}(fS^{r-l+j})]_{j=0}^{l-1}, \\ H_0 &= U_0^* M_t U_0, \quad H_f = U_f^* M_t U_f, \quad C = U_0^* M_t U_f. \end{aligned} \quad (\text{CTR8})$$

Each U has rank l . For U_0 , a polynomial of degree less than l belonging to $\mathcal{I}$ must be zero. For U_f , divisibility by f reduces a dependence in $\mathcal{I}$ to a polynomial supported in degrees $r-l, \dots, r-1$ and at the same time of degree at most $r-l-1$. Their ranges are $\mathcal{P} \cap \mathcal{I}^{\perp M_t}$ and $\mathcal{F} \cap \mathcal{I}^{\perp M_t}$, respectively. Thus

$$P_0 = P_I + U_0 H_0^{-1} U_0^* M_t, \quad P_f = P_I + U_f H_f^{-1} U_f^* M_t. \quad (\text{CTR9})$$

These are orthogonal decompositions inside each individual space; the cross Gram C between the two residual spaces remains present. By CTR5, the frame $B = [C_I, U_0, U_f]$ is invertible and satisfies

$$B^* M_t B = \begin{pmatrix} C_I^* M_t C_I & 0 & 0 \\ 0 & H_0 & C \\ 0 & C^* & H_f \end{pmatrix}, \quad B^{-1} D_t B = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -I_l & -H_0^{-1} C \\ 0 & H_f^{-1} C^* & I_l \end{pmatrix}. \quad (\text{CTR10})$$

Both formulas follow by applying CTR9 separately to its three displayed column groups; the second keeps both off-diagonal blocks with their signs.

Here is an exact spectral frame for this same endomorphism. Take the positive roots of H_0, H_f and put $T = H_0^{-1/2} C H_f^{-1/2}$. Positive definiteness of the lower Gram in CTR10 implies $I - T^* T > 0$, by its Schur complement. Choose orthonormal singular vectors a_j, b_j with $a_i^* T b_j = \sigma_j \delta_{ij}$ and $0 \leq \sigma_j < 1$. They can be constructed by diagonalizing the Hermitian matrix $T^* T$: for a positive eigenvalue σ_j^2 , set $a_j = T b_j / \sigma_j$; complete the resulting left vectors by an orthonormal basis of $\ker T^*$ at the zero singular values. This gives both orthonormal bases and the stated equation. The finite Hermitian spectral theorem itself follows by choosing one eigenvector, noting that its eigenvalue is real and its orthogonal complement is invariant, and then inducting on the dimension. This also constructs the positive square roots above.

Set, in the original coefficient space,

$$u_j = U_0 H_0^{-1/2} a_j, \quad v_j = U_f H_f^{-1/2} b_j, \quad s_j = (1 - \sigma_j^2)^{1/2} > 0, \quad w_j = \frac{v_j - \sigma_j u_j}{s_j}. \quad (\text{CTR11})$$

Their actual inner products are $u_i^* M_t u_j = v_i^* M_t v_j = \delta_{ij}$ and $u_i^* M_t v_j = \sigma_j \delta_{ij}$; hence $(u_j, w_j)_{j=1}^l$ is an M_t -orthonormal frame of $\mathcal{I}^{\perp M_t}$. On its j th two-dimensional plane, CTR9 acts by

$$P_0 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad P_f = \begin{pmatrix} \sigma_j^2 & \sigma_j s_j \\ \sigma_j s_j & s_j^2 \end{pmatrix}, \quad D_t = \begin{pmatrix} -s_j^2 & \sigma_j s_j \\ \sigma_j s_j & s_j^2 \end{pmatrix}. \quad (\text{CTR12})$$

Write $\alpha_j = \sqrt{(1-s_j)/2}$ and $\beta_j = \sqrt{(1+s_j)/2}$. Direct multiplication gives the orthonormal eigenvectors

$$e_j^+ = \alpha_j u_j + \beta_j w_j, \quad e_j^- = \beta_j u_j - \alpha_j w_j, \quad D_t e_j^+ = s_j e_j^+, \quad D_t e_j^- = -s_j e_j^-. \quad (\text{CTR13})$$

This includes $\sigma_j = 0$, when $s_j = 1$; no division by σ_j is used in CTR11 or CTR13. On $\mathcal{I}$ both projectors are the identity and their difference is zero. Consequently the complete inertia, rank, and trace of the original-metric absolute value are

$$\text{inertia}(D_t) = (l, l, r-l), \quad \text{rank } D_t = 2l, \quad \text{spec } D_t \subset [-1, 1], \quad \text{Tr } |D_t| = 2 \sum_{j=1}^l s_j \leq 2l. \quad (\text{CTR14})$$

Here $|D_t|$ is defined by the displayed M_t -orthonormal eigenframe; equivalently it is the positive root of D_t^2 in that same metric. These are the signed spectral maps of the original two subspaces, with their actual cross Gram, rather than an arbitrary rank- $2l$ matrix.

9.3 The derivative of the full ratio and its integrated bound

Let $a = \text{ess inf}_w \psi$, $b = \text{ess sup}_w \psi$, and $\text{osc}_w \psi = b - a$. For $t \in [0, 1]$ differentiation under each finite Gram integral gives

$$(M'_t)_{i,j} = \int_{\mathcal{K}_\epsilon} \overline{(c + iQz)^i} (c + iQz)^j \psi(z) e^{t\psi(z)} dw(z), \quad A_t = M_t^{-1} M'_t. \quad (\text{CTR15})$$

To justify differentiation, extend t to $[-1, 2]$. The derivative integrand is bounded in absolute value by a fixed constant times $\|\psi\|_\infty e^{2\|\psi\|_\infty}$, integrable because w is finite and the polynomial coordinates are bounded. The difference quotients have the same domination by the mean value theorem. Dominated convergence proves the derivative and its continuity.

The matrix A_t is the compression of multiplication by ψ to the original polynomial image in $L^2(w_t)$. More precisely, if $\mathcal{E} : \mathbb{C}^n \rightarrow L^2(w_t)$ is coefficient evaluation, then $\mathcal{E}^* \mathcal{E} = M_t$, $\mathcal{E}^* M_\psi \mathcal{E} = M'_t$, and its coefficient compression is $M_t^{-1} \mathcal{E}^* M_\psi \mathcal{E} = A_t$. It satisfies $A_t^* M_t = M_t A_t = M'_t$ and, for every coefficient x ,

$$a x^* M_t x \leq x^* M_t A_t x = \int \psi(z) |\mathcal{E}x(z)|^2 dw_t(z) \leq b x^* M_t x. \quad (\text{CTR16})$$

The notation $aI \leq A_t \leq bI$ means exactly this original-metric quadratic-form inequality.

Jacobi's determinant formula, with the fixed columns in CTR2, now gives

$$\begin{aligned} \frac{d}{dt} \log \mathcal{R}_{w_t}(f) &= \text{Tr}(G_f^{-1} C_f^* M'_t C_f) - \text{Tr}(G_0^{-1} C_0^* M'_t C_0) \\ &= \text{Tr}((P_f - P_0) M_t^{-1} M'_t) = \text{Tr}(D_t A_t). \end{aligned} \quad (\text{CTR17})$$

For completeness, Jacobi's formula follows by multilinearity of the determinant, differentiating one column at a time, and then using $\text{adj}(G) = \det(G)G^{-1}$; all G here are invertible by CTR3. Cyclicity and $M_t M_t^{-1} = I$ prove the second equality in CTR17.

In the complete actual eigenframe CTR13,

$$\text{Tr}(D_t A_t) = \sum_{j=1}^l s_j ((e_j^+)^* M_t A_t e_j^+ - (e_j^-)^* M_t A_t e_j^-). \quad (\text{CTR18})$$

The intersection contributes zero. Each scalar in parentheses is the difference of two numbers in $[a, b]$ by CTR16; therefore

$$|\text{Tr}(D_t A_t)| \leq \frac{\text{Tr } |D_t|}{2} (b - a) \leq l \text{osc}_w \psi. \quad (\text{CTR19})$$

This also follows by centering A_t at $(a+b)I/2$ and using $\text{Tr } D_t = 0$; the signed scalar cancellation is exact. All G_t and their inverses vary continuously on the compact t interval, so CTR17 is a continuous scalar derivative. Integration yields the completed logarithmic transport estimate

$$\boxed{|\log \mathcal{R}_{e^{\psi}w}(f) - \log \mathcal{R}_w(f)| \leq l \text{osc}_w \psi \leq 2l \|\psi\|_{\infty}.} \quad (\text{CTR20})$$

If $l = 0$ and $f = 1$, the two inclusions coincide and their ratio is identically one, giving the zero right side directly.

9.4 Application to the full original arithmetic polynomial

Retain the original packet, including every multiplicity:

$$\begin{aligned} k &= 4l + 1, \quad l \geq 1, \quad m \geq 1, \quad e = 1 + k(m-1), \\ q &= e(k+1)^2, \quad c = k/2, \quad r \in \{q, q+1\}, \\ d &= \delta^2, \quad g = \gamma^2, \quad 0 < \delta < \frac{1}{2}, \quad \gamma > 2, \end{aligned} \quad (\text{CTR21})$$

$$\chi(c+x) = \prod_{a,b=0}^k [x - (2a-k)\delta - i(2b-k)\gamma]^e, \quad f(S) = \prod_{j=0}^{l-1} (S - c - 2j - \frac{1}{2}).$$

Use $Q = q$, $\epsilon = 2^{-10}$ and the same compact bulk $\mathcal{K}_\epsilon$. The concrete positive base measure, with its full source mass and Jacobian, is

$$dw_{\text{pow}}(z) = q^{2q+1} |z|^{2q} \sigma(qz) dz, \quad \sigma(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi}. \quad (\text{CTR22})$$

For each fixed original q , this is a finite measure with strictly positive continuous density on both compact intervals: the Gamma argument has positive real part, so the original Gamma factor is finite, continuous, and nonzero there. Thus every Gram required in CTR3 is positive definite for this actual source; no moment or support nondegeneracy is assumed without verification.

Put

$$\begin{aligned} a_k &= \frac{k(k+2)(d-g)}{3q}, \quad \tau_\epsilon = \frac{k^2(d+g)}{q^2\epsilon^2}, \quad \delta_\chi = \frac{q\tau_\epsilon^2}{2(1-\tau_\epsilon)}, \\ \psi_\chi(z) &= \log \frac{|\chi(c+iqz)|^2}{(q|z|)^{2q}}. \end{aligned} \quad (\text{CTR23})$$

The finite condition $k \geq 2048\sqrt{d+g}$ ensures $k\sqrt{d+g}/q \leq \epsilon/2$ and hence $\tau_\epsilon \leq 1/4$, because $q \geq k^2$ and $2/\epsilon = 2048$. It makes every factor of $\chi(c+iqz)$ nonzero on the bulk, so the logarithm in CTR23 is a bounded continuous real function there.

Here is the exact quadratic expansion needed for its oscillation. Set $\zeta_{ab} = (2a-k)\delta + i(2b-k)\gamma$. For $|y| > k\sqrt{d+g}$,

$$\log \frac{|\chi(c+iy)|^2}{|y|^{2q}} = 2e \text{Re} \sum_{a,b} \log(1 + i\zeta_{ab}/y) = \frac{p_2}{y^2} + E_\chi(y), \quad p_2 = \frac{qk(k+2)}{3}(d-g). \quad (\text{CTR24})$$

The Taylor series of $\log(1+u)$ is absolutely convergent at every displayed factor. Odd powers cancel under the exact permutation $(a,b) \mapsto (k-a, k-b)$. The second power is $e \text{Re} \sum \zeta_{ab}^2/y^2$. The elementary finite sums $\sum_{a=0}^k (2a-k) = 0$ and $\sum_{a=0}^k (2a-k)^2 = (k+1)k(k+2)/3$ follow by expanding and using $\sum a = k(k+1)/2$ and $\sum a^2 = k(k+1)(2k+1)/6$. They give the coefficient p_2 with its sign and multiplicity in CTR24. For the remaining even terms put $\tau = k^2(d+g)/y^2 < 1$. There are exactly q roots counted with multiplicity, so

$$|E_\chi(y)| \leq q \sum_{j=2}^{\infty} \frac{\tau^j}{j} \leq \frac{q\tau^2}{2(1-\tau)}. \quad (\text{CTR25})$$

The first inequality bounds each real root power by its modulus and keeps the cancellation of odd powers already proved; the second uses $1/j \leq 1/2$ for $j \geq 2$ and sums the geometric series. Substitution $y = qz$ therefore proves on the whole bulk

$$\psi_\chi(z) = \frac{a_k}{z^2} + E_\chi(qz), \quad |E_\chi(qz)| \leq \delta_\chi, \quad \text{osc } \psi_\chi \leq |a_k|(\epsilon^{-2} - 64^{-2}) + 2\delta_\chi \leq 2(|a_k|\epsilon^{-2} + \delta_\chi). \quad (\text{CTR26})$$

For the first oscillation bound, z^{-2} lies in $[64^{-2}, \epsilon^{-2}]$ and the remainder's range lies in $[-\delta_\chi, \delta_\chi]$. This retains the quadratic signed term before bounding its oscillation.

The actual and quadratic-deformed measures are

$$\begin{aligned} dw_\chi &= q|\chi(c + iqz)|^2 \sigma(qz) dz = e^{\psi_\chi} dw_{\text{pow}}, \\ dw &= e^{a_k/z^2} dw_{\text{pow}}, \quad dw_\chi = e^{E_\chi(qz)} dw. \end{aligned} \quad (\text{CTR27})$$

Each measure has the verified positivity and finiteness of CTR22. Apply CTR20 with the same $\mathcal{P}$, same original f , and the same coordinate map $P(S) \mapsto P(c + iqz)$, to obtain

$$\boxed{\begin{aligned} |\log \mathcal{R}_{w_\chi}(f) - \log \mathcal{R}_{w_{\text{pow}}}(f)| &\leq l\{|a_k|(\epsilon^{-2} - 64^{-2}) + 2\delta_\chi\} \\ &\leq 2l(|a_k|\epsilon^{-2} + \delta_\chi), \\ |\log \mathcal{R}_{w_\chi}(f) - \log \mathcal{R}_w(f)| &\leq 2l\delta_\chi. \end{aligned}} \quad (\text{CTR28})$$

Both statements hold for $r = q$ and $r = q + 1$. Their coefficient is l , although each individual Gram determinant has r columns.

These are two forms on the same original relation coefficient space. Their exact realization inside the one bulk Hilbert space $L^2(\mathcal{K}_\epsilon, q\sigma(qz)dz)$ is

$$\begin{aligned} T_{\text{pow}}P(z) &= (iqz)^q P(c + iqz), \quad T_\chi P(z) = \chi(c + iqz)P(c + iqz), \\ (Uh)(z) &= \frac{\chi(c + iqz)}{(iqz)^q} h(z), \quad UT_{\text{pow}} = T_\chi, \quad UM_{f(c+iqz)} = M_{f(c+iqz)}U. \end{aligned} \quad (\text{CTR29})$$

The multiplier and its inverse are bounded on the compact bulk because their denominators and numerators are nonzero there. Its squared modulus is exactly e^{ψ_χ} . Taking the Hilbert Gram of these maps yields CTR27, including the numerator maps obtained by composing with the same f . Thus CTR29 proves the source-form comparison without introducing any replacement for the original arithmetic quotient $E_\chi = \mathbb{C}[S]/(\chi)$.

If monic z coordinates are used later, the exact polynomial is $\tilde{f}(z) = (iq)^{-l}f(c + iqz)$, and the coefficient change $P(S) \mapsto P(c + iqz)$ is triangular with diagonal $(iq)^j$, $0 \leq j < r$. Its common squared determinant cancels between the two restriction Grams, whereas the leading factor of f gives

$$\mathcal{R}_w^S(f) = q^{2lr} \mathcal{R}_w^z(\tilde{f}), \quad \log \mathcal{R}_w^S(f) = 2lr \log q + \log \mathcal{R}_w^z(\tilde{f}). \quad (\text{CTR30})$$

Indeed each of the r numerator polynomial columns acquires the factor $(iq)^l$, and its Gram determinant acquires $|(iq)^l|^{2r} = q^{2lr}$. This factor is retained separately in each source form before their logarithmic difference is taken.

Finally the bounds in CTR28 have explicit original-degree rates. On the displayed threshold $\tau_\epsilon \leq 1/4$,

$$|a_k| \leq \frac{|d - g|}{3e}, \quad \delta_\chi \leq \frac{2k^4(d + g)^2}{3e^3(k + 1)^6\epsilon^4} \leq \frac{2(d + g)^2}{3e^3k^2\epsilon^4}. \quad (\text{CTR31})$$

The first inequality uses $k(k + 2) < (k + 1)^2$. The second substitutes the original $q = e(k + 1)^2$ into CTR23 and uses $1/(1 - \tau_\epsilon) \leq 4/3$. Since $2l = (k - 1)/2$, the coarser first error in CTR28 is bounded by

$$\frac{(k - 1)|d - g|}{6e\epsilon^2} + \frac{(k - 1)(d + g)^2}{3e^3k^2\epsilon^4}. \quad (\text{CTR32})$$

For fixed δ, γ, m and fixed $\epsilon = 2^{-10}$ this is $O(k/e)$, hence $O(k) = o(q)$ because $q \geq (k+1)^2$. For fixed $m \geq 2$, $e \geq k(m-1)$, so the same explicit bound is even $O(1)$. The quadratic-retaining second error is bounded by

$$2l\delta_\chi \leq \frac{(k-1)(d+g)^2}{3e^3k^2\epsilon^4}, \quad (\text{CTR33})$$

which is $O(k^{-1})$ at $m = 1$ and $O(k^{-4})$ for every fixed $m \geq 2$. These are bounds on the complete restricted relation-Gram ratio with its original multiplication by f . The lower and far-tail source contributions remain in their separate full-source comparison; no bulk estimate is substituted for those omitted integrals.

9.5 Complete return to the original sigma shifted-moment determinants

We now transport the entire line integral, using the proved original inner and far estimates and retaining the exact four tail corrections. For $s = 0, l$ and $0 \leq i, j < r$, define the full real moment matrices

$$(\mathbf{M}_{q+s,r})_{i,j} = \int_{\mathbb{R}} y^{2(q+s)+i+j} \sigma(y) dy, \quad \mathcal{H}_{q,l,r} = \frac{\det \mathbf{M}_{q+l,r}}{\det \mathbf{M}_{q,r}}. \quad (\text{CTR34})$$

All entries are finite by the original Gamma exponential bound stated below. These Grams are positive definite: their weights $y^{2(q+s)}\sigma(y)$ are positive away from the single point $y = 0$, and a nonzero polynomial is nonzero on an interval away from that point. The moment ratio has therefore been specified by complete convergent integrals; its value is not assumed.

The original full relation Grams and two reference source forms, all on the same coefficient space $\mathcal{P}$, are

$$\begin{aligned} H_{\chi,0} &= \text{Gram}_\sigma[\chi S^j]_{j=0}^{r-1}, & H_{\chi,f} &= \text{Gram}_\sigma[f\chi S^j]_{j=0}^{r-1}, \\ H_{\text{pow},s} &= \left(\int_{\mathbb{R}} y^{2(q+s)} \overline{(c+iy)^i} (c+iy)^j \sigma(y) dy \right)_{i,j=0}^{r-1}, & s &= 0, l. \end{aligned} \quad (\text{CTR35})$$

In the first row every polynomial is evaluated at $S = c + iy$. The coefficient map $P(S) \mapsto P(c + iy)$ is $J_{h,j} = \binom{j}{h} c^{j-h} (\sqrt{-1})^h$ for $h \leq j$, with zero entries for $h > j$. Its diagonal entries are $(\sqrt{-1})^h$, so $|\det J| = 1$, and

$$H_{\text{pow},s} = J^* \mathbf{M}_{q+s,r} J, \quad \frac{\det H_{\text{pow},l}}{\det H_{\text{pow},0}} = \mathcal{H}_{q,l,r}. \quad (\text{CTR36})$$

This proves the exact translation and phase transport to the real Hankel matrices, with every coefficient map retained.

Partition the original line as $I = \{|y| < q\epsilon\}$, $B = \{q\epsilon \leq |y| \leq 64q\}$, $F = \{|y| > 64q\}$. For any one of the four positive full forms in CTR35, write $H = H^I + H^B + H^F$, and retain the exact positive correction

$$t_H = \log \det \left(I_r + (H^B)^{-1/2} (H^I + H^F) (H^B)^{-1/2} \right) \geq 0, \quad \log \det H = \log \det H^B + t_H. \quad (\text{CTR37})$$

The equality follows by factoring H by the two square roots of H^B . No tail integral has been discarded.

Here are the finite common bounds for these four corrections. Put

$$\begin{aligned} \kappa_{I,r} &= \frac{2\epsilon r^2}{\sqrt{\cos 1}} \left(\frac{4}{3} 2^{-13} e^{2\pi} \right)^q < \frac{2\epsilon r^2}{\sqrt{\cos 1}} e^{-2q}, \\ \kappa_F &= \frac{512q}{59\sqrt{\cos 1}} e^{-19q}, \quad L_r = r \log(1 + \kappa_{I,r} + \kappa_F). \end{aligned} \quad (\text{CTR38})$$

For $H_{\chi,0}, H_{\chi,f}$ the complete Gamma/Legendre calculation BRD22–BRD38 proves $H^I \preceq \kappa_{I,r} H^B$ and $H^F \preceq \kappa_F H^B$ on exactly the threshold in CTR23. We verify that the same constants apply to the two explicit reference forms, rather than transferring a theorem about a different quotient.

The two source bounds proved by the Euler contour rotation and reflection calculation BRD23–BRD24 are

$$C_L e^{-\pi|y|} \leq \sigma(y) \leq C_U e^{-|y|}, \quad C_L = \frac{\pi}{\Gamma(3/4)^2}, \quad C_U = \frac{\Gamma(1/4)^2}{2\pi\sqrt{\cos 1}}, \quad \frac{C_U}{C_L} = \frac{1}{\sqrt{\cos 1}}. \quad (\text{CTR39})$$

Their full derivation is in the accompanying BRD source. In particular these are bounds for the same unscaled sigma density as CTR34. The Rodrigues/orthogonality calculation BRD25–BRD26 gives, for any polynomial Q of degree at most $r-1 \leq q$,

$$|Q(z)|^2 \leq \sum_{j=0}^{r-1} (2j+1) |P_j(2z-3)|^2 \int_1^2 |Q(v)|^2 dv. \quad (\text{CTR40})$$

The P_j are the original Legendre polynomials on $[-1, 1]$. For $|z| \leq \epsilon$ their sum is at most $r^2 8^{2q}$; for $|z| \geq 64$ it is at most $r^2((4 + 6/64)|z|)^{2r-2}$. Indeed their elementary real exterior bound is $|P_j(x)| \leq (2|x|)^j$; substitute $x = 2z - 3$ and sum $\sum_{j < r} (2j+1) = r^2$.

For the reference form $H_{\text{pow},s}$, setting $y = qz$ retains the density factor $q^{2q+2s+1}|z|^{2q+2s}\sigma(qz)$. On $[1, 2]$ its quadratic form is at least

$$q^{2q+2s+1} C_L e^{-2\pi q} \int_1^2 |Q(z)|^2 dz. \quad (\text{CTR41})$$

On the inner region the numerator is bounded by $q^{2q+2s+1} \epsilon^{2q+2s} C_U r^2 8^{2q}$ times $2\epsilon \int_1^2 |Q|^2$. Dividing by CTR41 proves

$$H_{\text{pow},s}^I \preceq \frac{2\epsilon r^2}{\sqrt{\cos 1}} \epsilon^{2s} (2^{-14} e^{2\pi})^q H_{\text{pow},s}^B \preceq \kappa_{I,r} H_{\text{pow},s}^B. \quad (\text{CTR42})$$

Here $\epsilon = 2^{-10}$ makes $\epsilon^2 8^2 = 2^{-14}$, $\epsilon^{2s} \leq 1$, and $2^{-14} < (4/3)2^{-13}$. This is the reference's actual inner integral estimate at zero root radii.

For its far region, put $D_s = 2q + 2s + 2r - 2 \leq 5q$. The source and polynomial bounds in CTR39–CTR40, divided by CTR41, give

$$H_{\text{pow},s}^F \preceq \frac{2r^2}{\sqrt{\cos 1}} \frac{e^{2\pi q} (131/32)^{2r-2} 64^{D_s} e^{-64q}}{q - D_s/64} H_{\text{pow},s}^B. \quad (\text{CTR43})$$

To verify the integration constant, for $z \geq 64$ use $\log(z/64) \leq (z-64)/64$, which gives $\int_{64}^\infty z^{D_s} e^{-qz} dz \leq 64^{D_s} e^{-64q} / (q - D_s/64)$. The factor two counts both real tails. Since $q - D_s/64 \geq 59q/64$ and $64 - 2\pi - 5\log 64 - 2\log(131/32) > 20$, the coefficient in CTR43 is at most $128r^2 e^{-20q} / (59q\sqrt{\cos 1}) \leq \kappa_F$, using $r \leq q+1 \leq 2q$. Thus all four actual corrections in CTR37 obey

$$0 \leq t_H \leq L_r. \quad (\text{CTR44})$$

This follows by congruence, since every eigenvalue of its positive tail matrix is at most $\kappa_{I,r} + \kappa_F$.

On the common bulk, preserve f while applying CTR28. Define

$$E_{\chi,l} = l\{|a_k|(\epsilon^{-2} - 64^{-2}) + 2\delta_\chi\}, \quad \Delta_f = \frac{S_2}{q^2 \epsilon^2}, \quad S_2 = \sum_{j=0}^{l-1} (2j + \frac{1}{2})^2 = \frac{l(16l^2 - 12l - 1)}{12}. \quad (\text{CTR45})$$

The signed bulk source difference $d_\chi = \log \mathcal{R}_{w_\chi}(f) - \log \mathcal{R}_{w_{\text{pow}}}(f)$ satisfies $|d_\chi| \leq E_{\chi,l}$. For the remaining multiplication factor, at every bulk point

$$1 \leq \frac{|f(c+iy)|^2}{|y|^{2l}} = \prod_{j < l} (1 + t_j^2/y^2) \leq e^{\Delta_f}. \quad (\text{CTR46})$$

The upper bound follows by $\log(1+x) \leq x$ and $|y| \geq q\epsilon$. Integrating every squared polynomial gives the positive-form comparison between the bulk numerator with f and the bulk reference

numerator with $|y|^{2l}$. Congruence and multiplication of its r eigenvalues therefore give a one-sided bulk correction d_f satisfying $0 \leq d_f \leq r\Delta_f$. Its denominator is the identical $H_{\text{pow},0}^B$.

Writing the full original relation determinant as $\mathfrak{R}_{q+r-1} = \det H_{\chi,f} / \det H_{\chi,0}$, CTR36–CTR37 now prove the complete exact identity

$$\log \mathfrak{R}_{q+r-1} - \log \mathcal{H}_{q,l,r} = d_\chi + d_f + t_{H_{\chi,f}} - t_{H_{\chi,0}} - t_{H_{\text{pow},l}} + t_{H_{\text{pow},0}}. \quad (\text{CTR47})$$

All four original or reference nonnegative tail corrections remain separately in this formula. In particular its proved finite enclosure is

$$-E_{\chi,l} - 2L_r \leq \log \mathfrak{R}_{q+r-1} - \log \frac{\det \mathbf{M}_{q+l,r}}{\det \mathbf{M}_{q,r}} \leq E_{\chi,l} + r\Delta_f + 2L_r, \quad r = q, q+1. \quad (\text{CTR48})$$

The one-sided $r\Delta_f$ appears only in the upper endpoint. Equations CTR31–CTR33 bound $E_{\chi,l} = O(k/e)$. Also $S_2 \leq 4l^3 \leq k^3/16$ and $r \leq 2q$ imply $r\Delta_f \leq k^3/(8q\epsilon^2) \leq k/(8\epsilon\epsilon^2)$. Finally $L_r \leq r(\kappa_{I,r} + \kappa_F)$ decays exponentially in q times an explicit polynomial by CTR38. It follows from these displayed constants that

$$\log \mathfrak{R}_{q+r-1} = \log \frac{\det \mathbf{M}_{q+l,r}}{\det \mathbf{M}_{q,r}} + O(k) = \log \frac{\det \mathbf{M}_{q+l,r}}{\det \mathbf{M}_{q,r}} + o(q) \quad (\text{CTR49})$$

for each fixed original packet and m , and the error is $O(1)$ when the fixed multiplicity satisfies $m \geq 2$. The reference now depends only on the explicit integers q, l, r ; δ, γ remain in the proved error constants and threshold.

9.6 The full reference entries are exact sigma moments

To give a complete specification without unevaluated moment inputs, write $\mu_j = \int_{\mathbb{R}} y^j \sigma(y) dy$. The characteristic function JSA4 of the same original sigma density is

$$\int_{\mathbb{R}} e^{ity} \sigma(y) dy = c_\sigma (\cosh t)^{-1/2}, \quad c_\sigma = \sqrt{2\pi}. \quad (\text{CTR50})$$

Its direct derivation can be retained here. In the product of the Euler integrals for $\Gamma(a + iy/2)$ and $\Gamma(a - iy/2)$, use $u + v = w$ and $\log(u/v) = 2x$. The Jacobian and the positive w integral give $|\Gamma(a + iy/2)|^2 = 2\Gamma(2a) \int_{\mathbb{R}} e^{iyx} (2 \cosh x)^{-2a} dx$. Fourier inversion gives $2\Gamma(2a)(2 \cosh t)^{-2a}$ for its density characteristic function. The function $(2 \cosh x)^{-2a}$ is smooth with integrable exponentially decaying derivatives; Fourier inversion also follows by Gaussian regularization and dominated convergence. At $a = 1/4$ the constant is $2\Gamma(1/2)2^{-1/2} = \sqrt{2\pi}$, proving CTR50.

The exponential upper bound in CTR39 permits differentiation of every finite order under the integral in CTR50: the derivative is dominated by $|y|^j C_U e^{-|y|}$. Thus

$$\mu_{2h+1} = 0, \quad \mu_{2h} = c_\sigma (-1)^h \frac{d^{2h}}{dt^{2h}} (\cosh t)^{-1/2} \Big|_{t=0}, \quad (\mathbf{M}_{a,r})_{i,j} = \mu_{2a+i+j}. \quad (\text{CTR51})$$

The odd moments vanish either by the even characteristic function or by the original even density. These formulas retain the unscaled variable y and the full Gamma mass.

For an explicit finite rational recurrence let $c_h = ((d/dt)^{2h} (\cosh t)^{-1/2})|_{t=0}$ and $c_0 = 1$. The derivative equation $h'(t) \cosh t + \frac{1}{2}h(t) \sinh t = 0$ for $h(t) = (\cosh t)^{-1/2}$ gives, by comparing its coefficient of t^{2n-1} ,

$$c_n = - \sum_{j=1}^{n-1} \binom{2n-1}{2j-1} c_j - \frac{1}{2} \sum_{j=0}^{n-1} \binom{2n-1}{2j} c_j \quad (n \geq 1). \quad (\text{CTR52})$$

In this comparison the coefficient of c_n is one; the remaining terms come respectively from $h' \cosh$ and $h \sinh / 2$. It proves that every c_n is rational and determines it in finitely many operations. For example $c_1 = -1/2$ and $c_2 = 7/4$. Hence $M_{a,r} = c_\sigma Q_{a,r}$, where the rational matrix has entry zero at odd $i + j$, and entry $(-1)^{a+(i+j)/2} c_{a+(i+j)/2}$ at even $i + j$. In particular

$$\det M_{a,r} = c_\sigma^r \det Q_{a,r} > 0, \quad \mathcal{H}_{q,l,r} = \frac{c_\sigma^r \det Q_{q+l,r}}{c_\sigma^r \det Q_{q,r}}. \quad (\text{CTR53})$$

The mass factors are displayed in both determinants before cancellation. Thus the new reference is an exact positive rational Hankel-determinant ratio, supplied by a finite recurrence, rather than an unspecified comparison measure.

The existing low-endpoint constant C_{rel} is retained unchanged in the original identity $\mathfrak{T}_k = C_{\text{rel}} + \log \mathfrak{R}_{2q-1} + \log \mathfrak{R}_{2q}$. Substitution of CTR47 gives

$$\mathfrak{T}_k = C_{\text{rel}} + \sum_{r=q}^{q+1} \log \frac{\det M_{q+l,r}}{\det M_{q,r}} + \varepsilon_q + \varepsilon_{q+1}, \quad (\text{CTR54})$$

where each ε_r has the exact four-tail decomposition CTR47 and the one-sided bounds CTR48. Their sum is $O(k) = o(q)$, and $O(1)$ for fixed $m \geq 2$. The incoming Gamma return is $R_k^0 - R_k^\sigma = -\mathfrak{T}_k$, so its entire expression has the opposite sign. The next calculation is the asymptotic evaluation of these specified universal Hankel determinants; no asymptotic value has been inserted in CTR49 or CTR54.

The full-source arithmetic receiving application

This standalone application is accompanied by the complete original JSR source, the complete BRD and CTR proofs, and the full earlier providers. Its BHR1–15 body is also inserted, byte for byte, into the complete JSR successor. A cumulative compiler should print that body once, while retaining both complete source presentations.

10 The full original relation return through the universal Hankel source

Retain the entire original packet, $S = c + iy$, all original maps in JSR1–15, and put $d = \delta^2$, $g = \gamma^2$ and $e = e_k = 1 + k(m - 1)$, so $q = e(k + 1)^2$. The following full-source calculation applies on the explicit original domain

$$k = 4l + 1, \quad l \geq 1, \quad k \geq 2048\sqrt{d + g}, \quad \epsilon = 2^{-10}, \quad r \in \{q, q + 1\}. \quad (\text{BHR1})$$

Indeed $R_*/q = k\sqrt{d + g}/q \leq \epsilon/2$ and $t_{l-1}/q = (2l - 3/2)/q \leq \epsilon$ there. These are the BRD22 hypotheses, including a strict density expansion radius. Below this threshold the original exact jet, relation, and finite trace formulas remain in use.

The complete constants are

$$\begin{aligned} \alpha_k^\chi &= \frac{k(k + 2)(d - g)}{3q}, \quad \tau_\epsilon = \frac{k^2(d + g)}{q^2\epsilon^2} \leq \frac{1}{4}, \quad \delta_\chi = \frac{q\tau_\epsilon^2}{2(1 - \tau_\epsilon)}, \\ E_\chi &= l\{|\alpha_k^\chi|(\epsilon^{-2} - 64^{-2}) + 2\delta_\chi\}, \quad \Delta_f = \frac{S_2}{q^2\epsilon^2}, \quad S_2 = \sum_{j=0}^{l-1} (2j + \tfrac{1}{2})^2 = \frac{l(16l^2 - 12l - 1)}{12}, \\ \kappa_{I,r} &= \frac{2\epsilon r^2}{\sqrt{\cos 1}} \left(\frac{4}{3} 2^{-13} e^{2\pi} \right)^q, \quad \kappa_F = \frac{512q}{59\sqrt{\cos 1}} e^{-19q}, \quad L_r^{\text{tail}} = r \log(1 + \kappa_{I,r} + \kappa_F). \end{aligned} \quad (\text{BHR2})$$

Here α_k^χ is exactly CTR23's a_k , given a superscript to preserve its distinction from the arithmetic allowances a_k^-, a_k^+ in JSR20. Likewise L_r^{tail} is CTR38's L_r and is distinct from the finite trace endpoints. The complete BRD and CTR bodies accompany this calculation with these constants unchanged.

Let $H_{\chi,0}$ and $H_{\chi,f}$ be the original Grams of $(\chi S^j)_{j < r}$ and $(f\chi S^j)_{j < r}$, respectively. Their exact ratio is the GSR relation determinant R_{q+r-1} . The new full moment matrices are

$$(\mathbf{M}_{a,r})_{ij} = \int_{\mathbb{R}} y^{2a+i+j} \sigma(y) dy, \quad 0 \leq i, j < r, \quad \rho_r = \log \frac{\det \mathbf{M}_{q+l,r}}{\det \mathbf{M}_{q,r}}. \quad (\text{BHR3})$$

They are positive definite: the scalar weight is positive away from zero and a nonzero polynomial cannot vanish on an interval. Their finite moments retain the unscaled sigma density of mass $\sqrt{2\pi}$. The exact coefficient map $P(S) \mapsto P(c + iy)$ is

$$J_{hj} = \binom{j}{h} c^{j-h} (\sqrt{-1})^h, \quad 0 \leq h \leq j < r, \quad |\det J| = 1. \quad (\text{BHR4})$$

Thus the full reference source form for $|y|^{2(q+s)} |P(c + iy)|^2 \sigma(y) dy$ is $J^* \mathbf{M}_{q+s,r} J$, $s = 0, l$. This proves the precise coordinate morphism and its determinant cancellation. It introduces no additional $2lr \log q$ into the full- y ratio ρ_r .

Here is the full comparison, with the original tails and signs retained before estimation. Partition the line into $I = \{|y| < q\epsilon\}$, $B = \{q\epsilon \leq |y| \leq 64q\}$ and $F = \{|y| > 64q\}$. For any of the four original or reference forms H define

$$t_H = \log \det \left(I_r + (H^B)^{-1/2} (H^I + H^F) (H^B)^{-1/2} \right), \quad \log \det H = \log \det H^B + t_H. \quad (\text{BHR5})$$

BRD22–38 bounds the two original tail matrices, and CTR39–44 proves the same bounds for both reference matrices by their actual Gamma/Legendre integrals. Every eigenvalue of these positive matrices is at most $\kappa_{I,r} + \kappa_F$, so in each case $0 \leq t_H \leq L_r^{\text{tail}}$.

For clarity, the degree- l density transport concerns the original polynomial spaces $\mathcal{P} = \mathbb{C}[S]_{\leq r-1}$, $f\mathcal{P}$, inside $\mathbb{C}[S]_{\leq r+l-1}$. Their intersection is exactly $f\mathbb{C}[S]_{\leq r-l-1}$, by division by the original monic f . Its kernel and quotient maps are those in the complete exact sequence CTR4. On the bulk, the density logarithm for replacing $|y|^{2q}$ by $|\chi(c+iy)|^2$ is

$$\psi_\chi(z) = \alpha_k^\chi/z^2 + E_\chi^{\text{dens}}(qz), \quad |E_\chi^{\text{dens}}(qz)| \leq \delta_\chi, \quad \epsilon \leq |z| \leq 64. \quad (\text{BHR6})$$

This is the absolutely convergent expansion of the full root product, with the odd power sums cancelled by the original root permutation. Its second coefficient is $qk(k+2)(d-g)/3$ before the substitution $y = qz$. The difference of the two actual orthogonal projectors has inertia $(l, l, r-l)$, trace zero and trace norm at most $2l$; CTR8–14 constructs its full residual frame with its original cross Gram. Differentiating the two restriction determinants along $e^{t\psi_\chi}$ gives the trace of this projector difference against the compressed multiplication by ψ_χ . In its signed eigenframe, each paired expectation differs by at most $\text{osc } \psi_\chi$ and the positive weights sum to at most l . Integration therefore gives the actual bulk-ratio error d_χ with $|d_\chi| \leq l \text{osc } \psi_\chi \leq E_\chi$. All source coordinates, fixed inclusions and density Jacobians in this application are exactly CTR22–29.

The remaining numerator factor satisfies on the bulk $1 \leq |f(c+iy)|^2/|y|^{2l} \leq e^{\Delta_f}$. Integration on each original polynomial and multiplication of the r relative eigenvalues therefore gives a one-sided correction $0 \leq d_f \leq r\Delta_f$. Denote the reference forms from BHR4 by $H_{\text{pow},s}$. The determinant factorization BHR5 now proves

$$\begin{aligned} e_r &:= \log R_{q+r-1} - \rho_r = d_\chi + d_f + t_{H_{\chi,f}} - t_{H_{\chi,0}} \\ &\quad - t_{H_{\text{pow},1}} + t_{H_{\text{pow},0}}, \\ -E_\chi - 2L_r^{\text{tail}} &\leq e_r \leq E_\chi + r\Delta_f + 2L_r^{\text{tail}}. \end{aligned} \quad (\text{BHR7})$$

In particular the two numerator-density constants and all four nonnegative full-line tail corrections have their exact places in this equality. The upper-only $r\Delta_f$ has not been reflected into its lower side. This is the finite full-source CTR47–48 result returned through the exact original relation injection.

Its quantitative error is $O(k/e) + o(1) = o(q)$ on the original degree. Indeed CTR31 gives

$$|\alpha_k^\chi| \leq \frac{|d-g|}{3e}, \quad \delta_\chi \leq \frac{2(d+g)^2}{3e^3k^2\epsilon^4}, \quad r\Delta_f \leq \frac{k}{8e\epsilon^2}, \quad (\text{BHR8})$$

using $r \leq 2q$ and $S_2 \leq k^3/16$. The tail bound is at most $r(\kappa_{I,r} + \kappa_F)$, exponentially small in q times its displayed polynomial. Thus the source error is $O(k)$ for $m = 1$ and $O(1)$ for fixed $m \geq 2$. Every bound applies to the full original forms, not only their restriction to the bulk.

The full moment inputs in BHR3 are also exact. Retain $c_\sigma = \sqrt{2\pi}$ and let $c_0^{\text{mom}} = 1$,

$$c_n^{\text{mom}} = -\sum_{j=1}^{n-1} \binom{2n-1}{2j-1} c_j^{\text{mom}} - \frac{1}{2} \sum_{j=0}^{n-1} \binom{2n-1}{2j} c_j^{\text{mom}}. \quad (\text{BHR9})$$

The Euler-integral/Fourier calculation CTR50 proves the original characteristic function $c_\sigma(\cosh t)^{-1/2}$. Its derivative equation $h' \cosh t + h \sinh t/2 = 0$ proves BHR9 by comparing the coefficient of t^{2n-1} . The Gamma exponential bound permits differentiation of every finite order under the integral. Hence $\mu_{2n+1} = 0$, $\mu_{2n} = c_\sigma(-1)^n c_n^{\text{mom}}$ and $(M_{a,r})_{ij} = \mu_{2a+i+j}$. Every c_n^{mom} is rational by the finite recurrence. Thus each ratio in BHR3 is the logarithm of a positive rational determinant ratio, with c_σ^r present in both original determinants before cancellation.

The original low endpoints remain exactly $R_{q-1} = 1$ and $R_q = \mathfrak{r}_\chi$ with its JSR24b coefficient formula. Define the same complete low constant and the full-line reference centre by

$$\begin{aligned} C_{\text{rel}} &= \log D_{l,q-1} + \log D_{l,q} - \log \mathfrak{r}_\chi - \log D_{l,2q-1} - \log D_{l,2q}, \\ H_k^* &= -C_{\text{rel}} - \rho_q - \rho_{q+1}, \quad L_\Sigma = L_q^{\text{tail}} + L_{q+1}^{\text{tail}}, \\ b_k^- &= 2E_\chi + (2q+1)\Delta_f + 2L_\Sigma, \quad b_k^+ = 2E_\chi + 2L_\Sigma, \\ H_k^* - b_k^- &\leq \mathcal{H}_k = \mathcal{R}_k^0 - \mathcal{R}_k^\sigma = -\mathfrak{T}_k \leq H_k^* + b_k^+. \end{aligned} \tag{BHR10}$$

This is a direct signed substitution, not a different four-volume: JSR15 gives $\mathcal{H}_k = H_k^* - e_q - e_{q+1}$. Use the two upper sides of BHR7 for its lower endpoint and the two lower sides for its upper endpoint. The numerator factor Δ_f therefore occurs only in b_k^- . Both $b_k^\pm = o(q)$ with the explicit $O(k)$ and $O(1)$ branches just proved.

For comparison the prior BRD bulk calculation remains an additional proved interval. With

$$\begin{aligned} (K_{s,r})_{ij} &= \int_{\epsilon \leq |z| \leq 64} z^{i+j} |z|^{2q+2s} e^{\alpha_k^\chi / z^2} \sigma(qz) dz, \\ H_k^B &= -C_{\text{rel}} - 2l(2q+1) \log q - \sum_{r=q}^{q+1} \log \frac{\det K_{l,r}}{\det K_{0,r}}, \\ b_{B,k}^- &= 2(2q+1)\delta_\chi + (2q+1)\Delta_f + L_\Sigma, \quad b_{B,k}^+ = 2(2q+1)\delta_\chi + L_\Sigma, \end{aligned} \tag{BHR11}$$

BRD41a gives $H_k^B - b_{B,k}^- \leq \mathcal{H}_k \leq H_k^B + b_{B,k}^+$. The proof uses the same two high relation ranks and the same reversal of all four signs. The explicit $2l(2q+1) \log q$ is present in BHR11 because its integrals use $y = qz$. It is already incorporated in the full- y determinants in BHR3 and must not be added to BHR10 a second time.

On BHR1 define the actual current receiving endpoints by the intersection of these two intervals with the exact-low trace endpoints of JSR24a:

$$\begin{aligned} L_k^{\text{cur}} &= \max\{\mathcal{L}_k, H_k^* - b_k^-, H_k^B - b_{B,k}^-\}, \\ U_k^{\text{cur}} &= \min\{\mathcal{U}_k, H_k^* + b_k^+, H_k^B + b_{B,k}^+\}, \\ L_k^{\text{cur}} &\leq \mathcal{H}_k \leq U_k^{\text{cur}}, \quad 0 \leq U_k^{\text{cur}} - L_k^{\text{cur}} \leq b_k^- + b_k^+ = o(q). \end{aligned} \tag{BHR12}$$

Each interval contains the very same scalar by the proved morphisms JSR8–15 and BHR4–7. Thus its maximum and minimum are ordered and preserve all original information. For tensor orders outside BHR1 define $L_k^{\text{cur}} = \mathcal{L}_k$ and $U_k^{\text{cur}} = \mathcal{U}_k$ instead; all original finite formulas remain valid there.

This gives the following literal arithmetic return, with the original source deficit quantities of JSR18–21:

$$\begin{aligned} \mathcal{B}_k^0 + L_k^{\text{cur}} - a_k^- &\leq \mathcal{B}_k^{\text{ar}} \leq \mathcal{B}_k^0 + U_k^{\text{cur}} + a_k^+, \\ \mathcal{B}_k^0 + L_k^{\text{cur}} - D_k^- &\leq \mathcal{B}_k^{\text{ar}} \leq \mathcal{B}_k^0 + U_k^{\text{cur}} + D_k^+, \quad 0 \leq a_k^\pm \leq D_k^\pm. \end{aligned} \tag{BHR13}$$

Add the original arithmetic interval to BHR12 and use $\mathcal{B}_k^{\text{ar}} = \mathcal{B}_k^0 + \mathcal{H}_k + \delta_k^\sigma$ to prove both lines. The full Hankel centre gives in particular

$$\begin{aligned} -b_k^- - a_k^- &\leq \Delta_k^\Gamma - H_k^* = \delta_k^\sigma - e_q - e_{q+1} \leq b_k^+ + a_k^+, \\ -(64m + 245) &\leq \liminf \frac{\Delta_k^\Gamma - H_k^*}{q \log k} \leq \limsup \frac{\Delta_k^\Gamma - H_k^*}{q \log k} \leq 128m + 490 \quad (m \geq 2), \\ \limsup \frac{(U_k^{\text{cur}} + a_k^+) - (L_k^{\text{cur}} - a_k^-)}{q \log k} &\leq 192m + 735 \quad (m \geq 2). \end{aligned} \tag{BHR14}$$

The same finite bounds hold with $a_k^\pm$ replaced by $D_k^\pm$. Since $b_k^\pm = o(q)$, they vanish after division by $q \log k$. The signed ODD limits in JSR21 then prove the second line. For the last, add the two

finite allowances and use BHR12; the limit of $(D_k^- + D_k^+)/(q \log k)$ is $192m + 735$. For $m = 1$, retain instead $(\Delta_k^\Gamma - H_k^*)/(kq) = O_h((\log k/k)^{5/7})$; the $O(k)$ Gamma comparison error divided by kq is $O(q^{-1})$, which is smaller on this actual degree. No multiple-zero constant is assigned to the simple-zero regime.

Finally the full original HC threshold returns as

$$\begin{aligned} \max\{0, \mathcal{B}_k^0 + L_k^{\text{cur}} - D_k^- - 4q \log(D_h k)\} &\leq \mathcal{R}_k \\ &\leq \mathcal{B}_k^0 + U_k^{\text{cur}} + D_k^+ - 4q \log(D_h k). \end{aligned} \tag{BHR15}$$

Subtract its unchanged threshold from BHR13 and use the original independent constraint $\mathcal{R}_k \geq 0$. This computes the full-source approximation error at $o(q)$ while retaining the complete low moment ratio, the universal Hankel determinants and the actual arithmetic defect. It assigns no asymptotic value to H_k^* and makes no RH endpoint assertion.